

SECP3843 - PPG CASE STUDY

RECOVERABLE ASSETS (RA) & INVENTORY RISK MANAGEMENT (IRM)
TECHNICAL REPORT

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CHAPTER 1

INTRODUCTION

1.1 Overview

The capacity to have lean and responsive supply chain in the contemporary industrial environment has become one of the key determinants of the financial wellbeing of a company. In the case of global manufacturing giants such as PPG, where the inventory of raw materials at the facility includes not only volatile solvents but also specialized pigments, a more sophisticated and integrated approach to information is required and not only through traditional bookkeeping. As Reis and Housley (2022) define it, switching to the automated data system enables organizations to leave the stage of mere data collection and transition to the stage of generating high-trust insights that lead to executive decision-making.

In order to solve the issue of inventory obsolescence, the system must be redesigned through the creation of an end-to-end analytics pipeline on the Microsoft Azure platform. It separates and purifies raw operational data into structured format to be used by a high-level business intelligence by deploying a multi-layered Medallion Architecture. Instead of using the traditional, manual reports, the solution is based on the identification of Recoverable Assets (RA) and Magna Carta (MC) risks. The end result is to have a visualization of the inventory health in real-time so that the organization is able to reduce financial waste and at the same time production requirements with regard to inventory are met without disruption.

1.2 Problem Background

The handling of chemical and industrial raw materials is a risky affair since products like pigments, resins have a short shelf-life, which is usually rigid. According to Silver, Pyke,

and Thomas (2016), the first inefficiency in the supply chain is the phenomenon of frozen capital, meaning that the company finances are frozen in the untapped inventory. It is found that a major visibility gap is currently experienced by PPG due to the fact that the inventory snapshots, production orders and sales information are currently stored in fragmented data silos. This is not integrated and as such, it is hardly possible to detect at-risk materials until it is too late when it is expired or the so-called dead stock.

This paper is focused on two risks. To start with, the availability of RA which are resources that have amounts that are beyond 12 months of the estimated usage. It is thought that non-detection of such excesses at the early stages leads to soaring cost of warehousing and high insurance premiums. Second, MC risk is a serious threat as far as finances are concerned; it is the materials whose consumption has not occurred at all in nine months or whose expiry dates have already elapsed. In professional accounting, these materials demand a 100% financial write-down as this directly decreases the net profit of the company.

Additionally, the lack of connection between inventory and sales records implies that in case there is a shortage of the materials, the company cannot know instantly which particular customer sales orders are to be postponed. These result in reactive firefighting culture in which relationships with clients are ruined because of unpredictable lead times. Therefore, it is urgently required to have a single system that would help fill these gaps and convert disjointed files into an active risk-management tool.

1.3 Problem Statement

Although raw operational data are available, the absence of an automated and unified analytical system at PPG cause problems as follows:

- **Inventory obsolescence:** Raw materials that have a limited shelf time usually go out of order before they can be used up, causing losses.

- **MC risk:** The materials that has not been used within a period of nine months or so will be written off financially by 100 percent, which in turn will affect the profitability.
- **Information silos:** There are six distinct datasets (materials, inventory, purchase orders, production orders, suppliers, sales orders) which can hardly be compared manually.
- **Reactive Management:** The traditional systems see the issues only when they are too late and they must respond to the shortages or excess inventory.
- **Inability to scale:** Manual processes are susceptible to errors and cannot be expanded to support the modern supply chain demands.

1.4 Project Objective

The following are quantifiable objectives that guide this project:

- To research and examine RA and MC business logic in the PPG scenario.
- To create and build a three-tier Medallion Architecture (Bronze, Silver, Gold) on Microsoft Azure.
- To have a Star Schema data model that is able to compute the total amounts of financial provisions.
- To view the visualization of inventory risks and affected sales orders with an interactive Power BI dashboard.

1.5 Project Scope

The project focused on six synthetic datasets that represent core supply chain operations which are:

- materials.csv
- inventory_snapshot.csv
- purchase_orders.csv
- production_orders.csv
- suppliers.csv

- sales_orders.csv

These datasets represent the PPG raw data and will be integrated. It is for enabling the cross-functional analysis of inventory behavior and demand. Furthermore, to create and build the three-tier Medallion Architecture on Microsoft Azure, by using it, Microsoft Azure as the main platform. The utilization of Azure Blob Storage is for the storage of raw and processed data. Meanwhile the process of data ingestion and orchestration is used by Azure Data Factory before being delivered to Power BI for visualization and reporting.

Plus, the system is designed to support some functionalities:

- Risk Detection
 - Identification of RA (excess inventory > 12 months)
 - Identification of MC risks (inactive or expired inventory)
- Inventory Monitoring
 - Detection of excess stock
 - Detection of expired materials
 - Detection of dead stock
- Financial Impact Analysis
 - Estimation of inventory holding costs
- Sales & Operations Impact
 - Identification of stockouts
 - Mapping shortages to affected customer sales orders
- Visualization
 - Development of an interactive dashboard to monitor inventory health in real time

1.6 Project Importance

To help reduce financial losses and improve operational efficiency in supply management, PPG should be alert by detecting RA and MC risk early. This can minimize inventory write-offs, reduce excess stock, and avoid frozen capital tied up in unused materials. The system transforms fragmented data into a unified analytics platform which allows management to monitor inventory health in real time and make faster decisions. This will reduce reliance on manual reporting that will lead to less human error that will improve overall efficiency in inventory planning and control.

The change from reactive problem-solving to proactive decision-making put risk management at its top. By integrating inventory, production, and sales data, the system helps ensure materials are available when needed. No overstocking and stockouts occur. The use of cloud technologies such as Microsoft Azure, Azure Data Factory, and Power BI also makes the system scalable and future-ready. This allow companies to expand their analytics capabilities.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Modern supply chains generate large amounts of data from inventory, production, and sales activities. Without a structured data architecture, these datasets remain fragmented and difficult to analyze. It will limit an organization's ability to detect inventory risks such as excess stock, dead stock, and stockouts. In the context of the PPG supply chain scenario, the transition from manual spreadsheet-based tracking to an automated analytics pipeline requires a strong theoretical foundation in three aspects. They are inventory management logic, financial provisioning principles, relational data modeling, and modern cloud data architecture. In addition, the review explores Medallion Data Architecture, cloud computing platforms such as Microsoft Azure, and modern data pipeline approaches such as ETL (Extract, Transform, Load) and ELT (Extract, Load, Transform). Together, these theories and technologies form the foundation for the design and implementation of the inventory risk analytics system proposed in this study.

2.2 Supply Chain Inventory Management

In the context of PPG supply chain, a firm grounding in inventory logic, accounting principles, and relational data modeling is highly required for the transition from manual tracking to automated data engineering.

2.2.1 Supply Chain Management in Inventory

Supply Chain Management (SCM) provides the essential framework for effective inventory practices, particularly in complex manufacturing environments like PPG. It involves the coordinated and integrated management of material, information, and financial flows throughout the entire supply chain from initial suppliers through production to final customers. The primary goal is to ensure the right materials are available in the right quantity, at the right time, and at the lowest possible cost, ultimately supporting operational efficiency. Core Principles of SCM in Inventory.

A fundamental principle of SCM is achieving an optimal balance between supply and demand.

- Excess Inventory: Leads to significant carrying costs (storage, insurance, capital lock-up).
- Insufficient Inventory: Causes costly stockouts, production delays, and failure to meet customer demand.

The SCM approach seeks to find the level that minimizes financial burden while maintaining seamless operations.

SCM is critical for integrating disparate operational datasets, transforming fragmented data into a cohesive system. This integration connects various stages of the supply chain:

- Upstream: Suppliers and Purchase Orders
- Internal Operations: Inventory Snapshots and Production Orders
- Downstream: Sales Orders (Customer Demand)

By connecting these components, SCM enables end-to-end visibility, allowing organizations to monitor the material flow lifecycle. This capability is vital for proactively identifying inefficiencies such as overstocking, understocking, or slow-moving stock.³ Inventory Visibility and Traceability

In the absence of a unified SCM system, data silos lead to inaccurate tracking and reactive management. Implementing a robust data engineering pipeline as highlighted in this project centralizes fragmented data. This transformation supports real-time monitoring and enables proactive decision-making by providing clear visibility into material usage and demand patterns.⁴ Risk Management

SCM places a strong emphasis on inventory risk management, which is especially important for time-sensitive or perishable materials. In the PPG context, key inventory risks identified and managed include:

- Excess Inventory (Recoverable Assets or RA): Materials exceeding 12 months of

expected usage.

- Obsolete/Inactive Inventory (Magna Carta or MC): Materials with no usage over 9 months or those that have expired.

Early detection of these risks, which directly impact financial and operational performance, allows for corrective actions such as reallocation, demand adjustment, or liquidation strategies. Conclusion

In essence, Supply Chain Management is the foundational structure that integrates all inventory data across business functions. It facilitates a crucial shift from fragmented, reactive inventory processes to a data-driven, proactive management approach. This modern approach is essential for reducing waste, enhancing service levels, and ultimately improving overall supply chain performance.

2.2.2 Inventory Management Logic

Modern inventory management aims for "lean" operations, balancing stock availability to meet demand with minimizing capital tied up. Within the PPG framework, this principle is managed using the concept of RA.

- A key rule for identifying RA is the 12-Month Consumption Rule: Any inventory quantity that surpasses the forecasted consumption for the next year is categorized as "Excess Stock."
- The Operational Impact of identifying RA is significant. Flagging these assets enables planners to intervene before materials expire. This triggers "recovery" actions, such as reallocating materials to other production lines or offering discounts for a quick sale. As noted by Silver et al. (2016), reducing this excess inventory is crucial for lowering "carrying costs," which encompass expenses like warehousing, insurance, and the opportunity cost associated with frozen capital.

2.2.3 Financial Provisioning

While excess stock is a nuisance, "Dead Stock" is a direct hit to the balance sheet. In the chemical industry, raw materials like resins and pigments are volatile and have strict shelf lives.

- **The MC Definition:** This is a high-risk category for materials that have shown zero consumption for 9 consecutive months or have already reached their expiration date.
- **100% Financial Write-down:** Under standard accounting "provisions," once a material hits the MC threshold, it is no longer considered an asset. The company must record a 100% financial write-down. This means the value of that inventory is effectively "erased" from the profit margins, acting as a total loss.
- **The Necessity of Early Warning:** Data engineering pipelines aim to transform this from a "post-mortem" realization into a predictive warning, allowing the company to avoid the "MC" death sentence through proactive inventory aging analysis.

2.3 Data Architecture

The Medallion Architecture is a data design pattern that organizes data into three progressive tiers of quality and structure to enable organizations to build reliable and scalable analytics systems. This architecture, originally formalized by Databricks (n.d.), establishes a clear separation between raw ingestion, validated transformation, and curated presentation of data. The purpose is to organize unstructured data lakes. It is also commonly known as "multi-hop" architecture. Within the context of the PPG inventory risk analytics project, the Medallion Architecture serves as the foundational framework for converting six fragmented operational CSV files into a governed dataset that supports the detection of RA and MC risks.

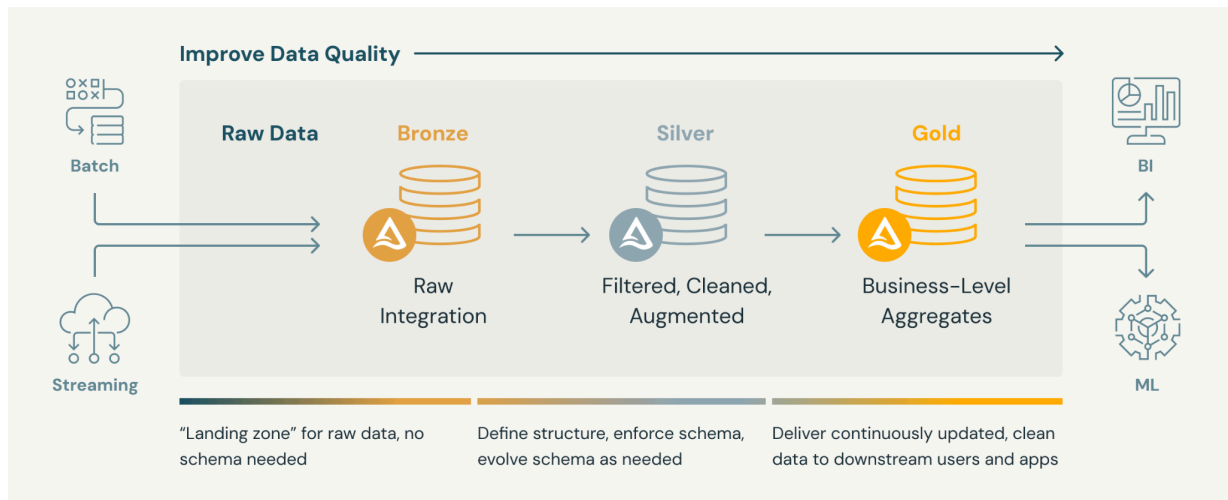


Figure 2.1. The Medallion Architecture (Databricks, n.d.)

As illustrated in Figure 2.1, the Medallion approach consists of three main layers:

- Bronze:** This layer serves as the initial landing zone for storing raw data from all source files. The data is ingested in its rawest form to preserve the original state and provide a complete audit trail. The defining characteristic of the Bronze layer is immutability because data stored in this tier is not altered or deleted. This supports the concept of a source of truth, as any errors in later stages can be traced back to the original data and reprocessed (Reis & Housley, 2022). The raw data remains available for recovery when data quality issues are found.
- Silver:** The Silver layer functions as the validation engine where the raw data is cleaned and normalized. This tier represents the transformation phase of the ELT (Extract, Load, Transform) paradigm, in which data is first loaded into the data lake before business logic is applied. Errors are corrected, and missing values are handled. Duplicate records are removed to improve data quality. The datasets are then combined to form a consistent structure.
- Gold:** The Gold layer represents the final stage in the pipeline, where the data is ready for business use. At this tier, Silver data is restructured into a Star schema. It organizes data into a central fact table and multiple dimension tables, which makes it easier to

aggregate and analyze using reporting and visualization tools such as Power BI. Specific logics are applied here to aggregate the refined data into actionable metrics for detecting RA and MC risks.

The Medallion Architecture improves data quality by enforcing validation at each layer before data advances further in the pipeline. Each stage of the architecture represents a new data accuracy and quality challenge. Data teams must ensure that issues such as missing values, schema mismatches, and duplicate records are detected and corrected before reaching the final analytical layer (Bergh, 2024).

2.4 Data Modeling

To turn fragmented CSV files into actionable insights, the data must be structured for analytical performance rather than just storage. Dimensional modeling, popularized by Kimball and Ross (2013), provides the foundational framework for this project. While a traditional Star Schema positions a single Fact Table at the center containing quantitative metrics surrounded by descriptive Dimension Tables, complex supply chains require a more advanced variation known as a Fact Constellation Schema. In this model, multiple fact tables share conformed dimension tables, allowing different business processes to be analyzed concurrently as shown in Figure 2.2.

This approach offers several key advantages tailored to the PPG inventory ecosystem. In terms of performance, it minimizes complex, resource-heavy joins, allowing tools like Microsoft Power BI to aggregate millions of rows efficiently. From a usability perspective, shared dimensions provide an intuitive structure for business users; for example, it makes it easier to understand how a stockout tracked in the inventory fact table directly impacts pending fulfillments in the sales order fact table via a shared "Material Dimension." Additionally, the Fact Constellation Schema ensures consistency by reducing data redundancy commonly found in flat files, so that updates such as changes to a supplier's metadata are reflected uniformly across the entire analytical dataset.

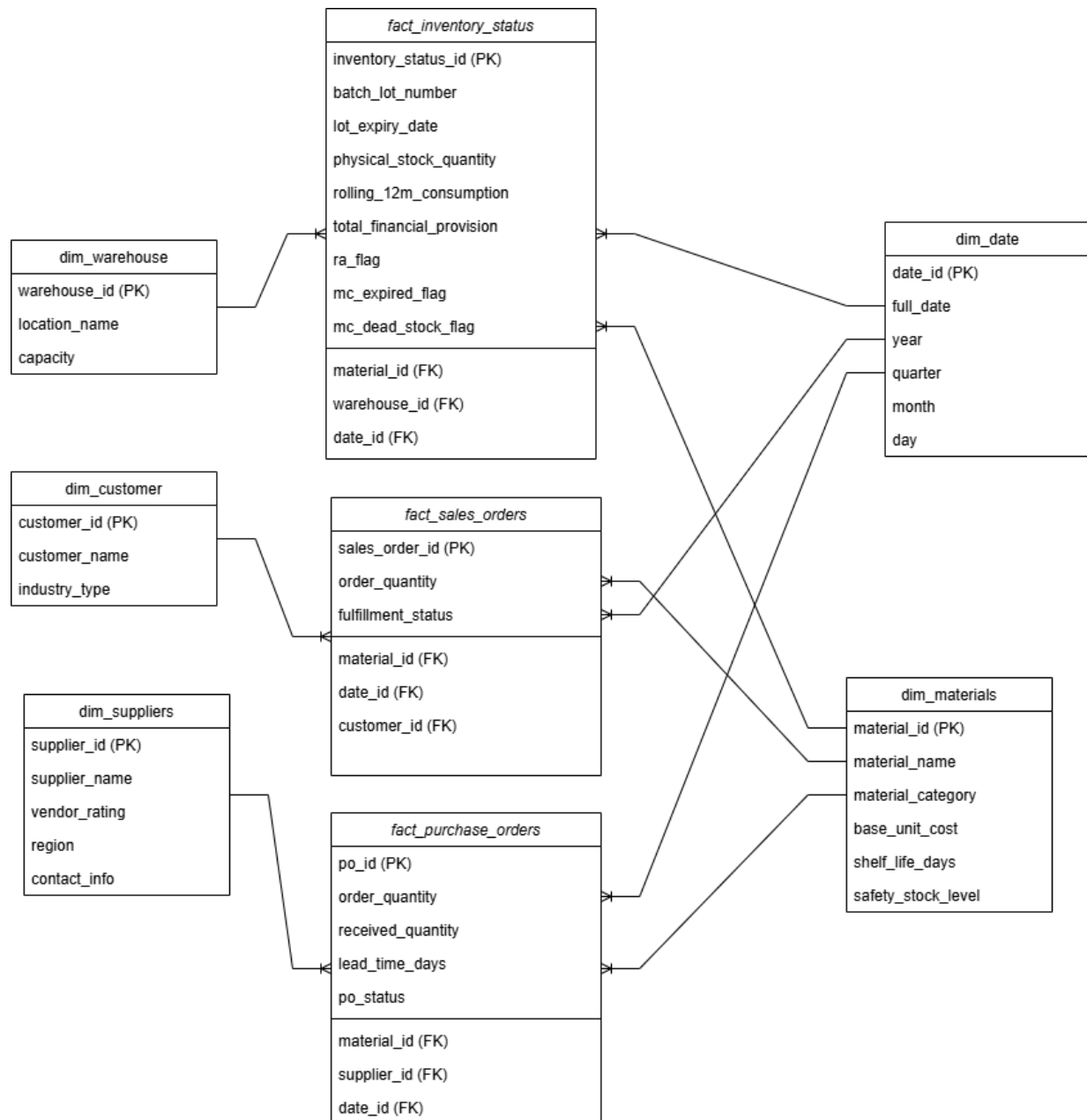


Figure 2.2. Facts Constellation Schema with Fact Tables and Dimension Tables

2.5 Enterprise Resource Planning (ERP)

As we know, Enterprise Resource Planning (ERP) refers to the technology used to integrate core business processes (Staff, 2026b). It gathers data from across the organization into one central database. That is why it is called “single source of truth”. In the PPG context,

the ERP is really needed for ensuring data consistency and operational coordination. All of the business transactions are recorded through ERP in real time.

However, ERP is not designed for advanced analytics. Relying only on ERP can cause issues such as excess inventory or dead stock. So, to overcome it, a data pipeline is required so that it can extract, transform, and load (ETL/ELT) this raw data into a structured format suitable for analysis. This pipeline enables the application of inventory management logic, such as the identification of excess (RA) and obsolete (MC) stock, ultimately supporting data-driven decision-making.

2.6 Technology Comparison

Selecting the most suitable technology is crucial as it provides most compatible services to manage complex inventory data and detect RA and MC risks automatically. This section will compare the traditional data management approach against modern cloud-based approach to justify the selected technology.

2.6.1 Cloud vs. On-Premise

The scalability and cost-efficiency are the factors that differentiates Microsoft Azure from traditional local SQL servers. On-premise SQL servers need an upfront capital expenditure (CapEx) for hardware, licensing and ongoing maintenance (Erl et al., 2013). In addition, organisations must over-provision hardware to support peak processing loads which leaves costly hardware as excess capacity during off-peak times. Cloud platforms like Microsoft Azure, on the other hand, offer dynamic elasticity and a more flexible operational expenditure (OpEx) model. It is immensely useful for inventory risk management systems because the required computing resources can be adjusted to accommodate the changing volume of data. This elasticity means that organisations only pay for services used, without the cost burden and inflexibility of on-premise solutions.

2.6.2 ETL vs. ELT

Traditional data pipeline would usually involve ETL pipeline where data is extracted, transformed in a separate server then loaded into the data warehouse. While it does work great for smaller datasets, that is not the case when massive data volumes come into play in modern-day data engineering. Therefore ELT pipelines like the ones represented in Figure 2.3 are desirable as they address the issue of transformation bottlenecks by leveraging the enormous processing capabilities of cloud data warehouses and data processing tools (Reis & Housley, 2022).

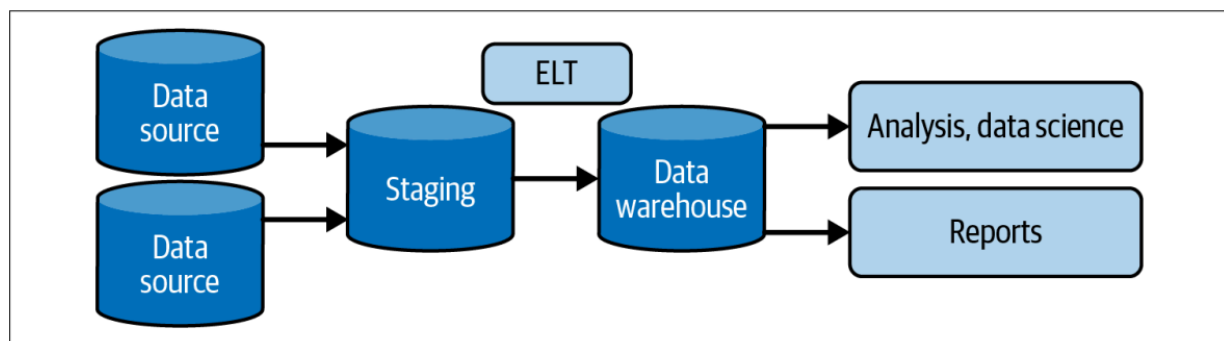


Figure 2.3. ELT Process Flow (Reis & Housley, 2022)

For the purpose of this case study, the raw CSV files will be extracted and loaded into a scalable Data Lake (the Bronze layer). They are then within cloud computing engines to form complex inventory metrics, specifically for identifying RA and MC risks. The end result of this overall ELT pipeline not only reduces processing bottlenecks when handling massive ERP CSV exports, but also maintains a historical catalogue of the raw inventory data for future finance reporting.

2.6.3 Visualization Tools

According to the traditional design, the inventory reports are prepared based on individual Microsoft Excel spreadsheets that have issues resulting from human errors when processing millions of rows of data. Modern solutions use data pipelines to feed data into data visualization tools like Microsoft Power BI. Power BI uses a compact in-memory engine with the Data Analysis Expressions (DAX) language to analyze large data sets that have complex relational database models (Microsoft, 2023). The system will use Microsoft Power BI and the DAX language to automate the inventory reporting process for RA and MC inventory. This new improvement will automate calculations that would otherwise be complicated for team members to manually calculate as well as create interactive data visualizations for stakeholders.

2.7 Related Work and Case Studies

In the manufacturing industry, data engineering pipelines have successfully reduced Days of Inventory on Hand (DIOH) through the integration of real-time data from supply chains, production systems, and inventory systems (BPCS, 2025). With the help of Azure-powered Inventory Control Towers and ELT pipelines, a global manufacturer reduced inventory by over 15%, improved forecasting accuracy by 20%, and saved over \$100 million annually by optimizing stock levels and minimizing excess holdings (BPCS, 2025). With real-time insights from integrated data pipelines, WNS' analytics-led model for a manufacturing major increased on-time-in-full (OTIF) deliveries by 5-10% and cut carrying costs (WNS, 2025).

Cloud-native tools like Microsoft Azure offer a proven industry approach. They handle complex data transformations and scalable ETL processing (Nalluru, 2024). Manufacturing enterprises like FRÄNKISCHE Industrial Pipes and Epiroc demonstrate this success. They use Azure Data Factory and cloud architecture to integrate scattered data. This process creates a unified view from hybrid sources.

Official Microsoft case studies highlight Epiroc's specific achievements. Epiroc established an AI and data factory on Azure. This platform standardized their operational processes globally. It significantly increased efficiency and reduced product waste (Microsoft, 2023). These companies actively moved away from manual coding. They leveraged Azure's scalable environment instead. This shift eliminated the latency found in traditional data integration methods.

The transition enables real-time, cloud-based data processing. It allows for the proactive management of inventory and stockout risks. Consequently, supply chain planners can reallocate RA before they expire.

Using the Medallion Architecture (Bronze-Silver-Gold layers on Azure Data Factory), these implementations ingest multi-source CSVs, apply RA/MC logic, and deliver a star schema for Power BI dashboards addressing DIOH directly through excess stock detection (RA: >12 months consumption) and obsolescence flagging (MC: 9+ months no use or expired) (Databricks, 2022; Bergh, 2024). With real-time insights from integrated data pipelines, ETL/ELT workflows support financial provisioning and sales order impact analysis through ERP data aggregation for demand forecasting (WNS, 2025; Reis & Housley, 2022).

2.8 Comparison with Previous Studies

To validate the proposed data engineering solution for PPG's inventory risk management, it is essential to compare the planned architecture against the established industry implementations reviewed in Section 2.5. Rather than deviating from industry norms, the proposed pipeline deliberately adopts the proven frameworks utilized by global manufacturing enterprises like Epiroc, BPCS, and WNS.

The following table summarizes how the proposed PPG project aligns with and implements the successful strategies identified in previous industry case studies:

Architectural / Business Component	Industry Implementations (BPCS, WNS, Epiroc)	Proposed PPG Analytics Pipeline
Data Processing Framework	Utilizes scalable ELT pipelines within cloud environments to handle massive datasets efficiently, often processing live ERP streams from hybrid on-premise and cloud sources (BPCS, 2025; WNS, 2025).	Extracts six synthetic CSV files into Azure Data Lake Storage Gen2 (Bronze), then applies ELT transformations using Azure Databricks. Scope is limited to batch CSV exports rather than live ERP streams, which is appropriate for a proof-of-concept context.
Architectural Design	Implements the full Medallion Architecture (Bronze, Silver, Gold) on Azure Data Factory, typically integrated with enterprise-grade monitoring and CI/CD pipelines (Databricks, 2022; Bergh, 2024).	Adopts the exact three-tier Medallion structure using Azure Data Factory for orchestration and Azure Databricks for processing.
Inventory Risk Logic	Applies RA (>12 months excess stock) and MC (9+ months inactivity or past expiry) logic to detect at-risk inventory and trigger financial write-downs (WNS, 2025; Databricks, 2022).	Automates the exact RA and MC business logic via Databricks to flag at-risk materials and compute 100% financial write-downs.

Visibility and Delivery	Deploys real-time Inventory Control Towers via Power BI for cross-functional, enterprise-wide supply chain visibility, often connecting to live data feeds (Microsoft, 2023; BPCS, 2025).	Delivers an interactive Power BI dashboard featuring DAX measures, KPI scorecards, and cross-filtering for sales order impacts.
Scale and Scope	Deployed globally across enterprise hybrid sources to achieve millions in cost savings, reduce product waste, and improve forecasting accuracy by 20% (BPCS, 2025; WNS, 2025).	Serves as a targeted, end-to-end proof of concept focused strictly on six core operational datasets to prove risk detection viability.

Table 2.1: Comparison of Proposed Pipeline with Previous Industry Studies

2.9 Microsoft Azure

The cloud platform selection influences the extent to which the source CSV files can successfully be converted to actionable insights. Microsoft Azure was chosen due to the fact that its services perfectly match the three layer Medallion Architecture. In contrast to generic cloud-based storage solutions, Azure offers a full ecosystem, where ingestion of raw data, transformation, and delivery to the dashboard can be managed under a single platform. This remove the intricacy of joining together services offered by various vendors. In addition, the project requires a platform capable of supporting batch processing of historical inventory snapshots as well as quick querying of the ultimate Power BI dashboard. Azure fulfills both requirements using its integrated service architecture (Rucco et al., 2024).

2.9.1 Map of Azure Services to PPG Pipeline Layers.

The Medallion Architecture is mapped to particular services of Azure. In the case of Bronze Layer, the six raw CSV files get stored in the Azure Data Lake Storage Gen2 as received, with the original data saved so that it can be audited (Carrilho, 2025). This imbalance is essential since in case any transformation logic requires reconfiguration, then the entire pipeline can be reprocessed using the original source of data. In the case of Silver layer, Azure Databricks uses the given PPG business logic. This involves the computation of 12-month rolling consumption of every material, the marking of materials whose current stocks are above the set limit as Recoverable Assets, and the recognition of Magna Carta materials. Azure Databricks can perform these window operations and date comparisons in a highly efficient manner on the entire dataset. In the case of Gold layer, the final star schema tables are stored in Azure SQL Database or Azure Synapse, then this structured format is directly linked to Power BI to visualise the dashboard.

2.9.2 Azure Data Factory to orchestrate Pipelines.

The project will demand the pipeline to execute in a synchronized order: ingest CSV files, run transformations, and load results into the Gold layer. This whole work process is coordinated by Azure Data Factory. In particular, Data Factory moves CSVs in their source locations to the Azure Data Lake Storage (Bronze), runs the Azure Databricks notebook with the RA and MC transformation code (Silver), and transfers transformed data to the Gold layers tables. This staging helps in making sure that the various layers are finished successfully before the next one gets started and also in case one step fails the pipeline can be restarted at the failure point as opposed to starting it up again.

2.9.3 Power BI Dashboard Delivery Integration

The end project deliverable will consist of a power BI dashboard to answer seven targeted questions such as what materials fall below the reorder levels and how customer sales orders are affected by stockouts. With native integration to Power BI, this can be done with a

lot of ease using Azure. After populating the Gold layer tables in the Azure SQL Database, power BI also connects to it in real-time with built-in connectors providing an incremental refresh. This implies that as new inventory data is fed, the dashboard is updated automatically. To the management team at PPG, this will mean real-time access to MC provisions and at-risk sales orders.

2.10 Microsoft Power BI

This project uses Microsoft Power BI, which is a self-service, unified business intelligence platform to develop the final analytical layer in inventory management. It is the main entry point of the stakeholders to the Gold layer of the Medallion Architecture.

2.10.1 Data Connectivity and Integration

The system employs the use of Microsoft Power BI to reach out to the data warehouse in Azure. Microsoft Power BI preserves the relational integrity achieved in the transformation stage by importing the Star Schema which includes the fact-inventory-status table and related dimensions (Materials, Suppliers, Date, etc.) into the database.

2.10.2 Data Analysis Expressions (DAX)

The project uses Data Analysis Expressions (DAX) to go beyond mere data viewing. The complex RA and MC business logic is automated by creating custom measures using DAX. The essential measurements that are computed using DAX are:

- **Total Financial Provision:** A dynamic value of flagged material (MC Dead Stock or Expired) value.
- **Inventory Risk Flags:** Indicators that assess the differences between inventory levels and reorder levels to detect stockout risks.

- **RA Identification:** Determining the materials that are over the rolling 12-month consumption threshold.

2.10.3 Interactive Visualization and Reporting

The last dashboard is to be an Inventory Control Tower. It offers real-time inventory health visibility by a number of essential visual components:

- **KPI Scorecards:** The value of MC provisions and at-risk sales order numbers in real-time.
- **Cross-Filtering Capabilities:** This will enable the user to pick a certain at-risk material and immediately view what sales orders of customers will be affected.
- **Drill-down Functionality:** This allows supply chain planners to navigate between a high-level category view (e.g., Resin, Pigment) to batches or lot expiry dates.

2.11 Summary

In order to build an effective inventory risk analytics system, it is important to have a strong foundation in inventory management logic, financial provisioning principles, data modeling, and cloud data architecture. The concepts of RA and MC risk demonstrate how insufficient inventory and dead stock can significantly impact operational efficiency and financial performance. Early identification of insufficient stock helps reduce carrying costs such as warehousing, insurance, and opportunity cost. Meanwhile, the identification of dead stock ensures that financial write-downs are properly recognized and managed. The star schema model enables efficient analytical processing and supports business intelligence reporting. Microsoft Azure provides scalability, flexibility, and processing power compared to traditional on-premise systems, while ELT pipelines allow large datasets to be processed efficiently within cloud environments. Visualization tools such as Power BI further support decision-making. It is by transforming complex inventory data into interactive dashboards and actionable insights. In the end, industry implementations demonstrate that integrating cloud

data pipelines, structured data models, and business intelligence tools can significantly reduce inventory costs, improve forecasting accuracy, and enhance supply chain visibility.

CHAPTER 3

METHODOLOGY

3.1 Introduction

Systematic methodology is critical in converting business needs into a working data analytics system. In the absence of well-defined methodology, data engineering projects may deliver pipelines that are hard to maintain, not well-documented, or do not meet the expectations of stakeholders. In this chapter, the methodology that will be used in the implementation of the inventory risk management solution is introduced.

The project uses the System Development Life Cycle (SDLC) as the general model. SDLC offers a framework of steps to follow in a project, starting with its conception, all the way to the final deployment and maintenance. Dennis, Wixom, and Tegarden (2015) argue that SDLC is especially appropriate with information systems since it has defined deliverables, milestone review, and quality control checkpoints in every phase. These features are suitable to the needs of this project, which implies numerous data sources, particular business logic to classify the inventory, and a final dashboard deliverables.

One of the benefits of SDLC in this project is its ability to support parallel work streams in the implementation phase. Rucco et al. (2024) note that cloud-based data engineering initiatives can enjoy the benefits of parallel development since services like storage, compute and visualisation can be set up separately and then integrated.

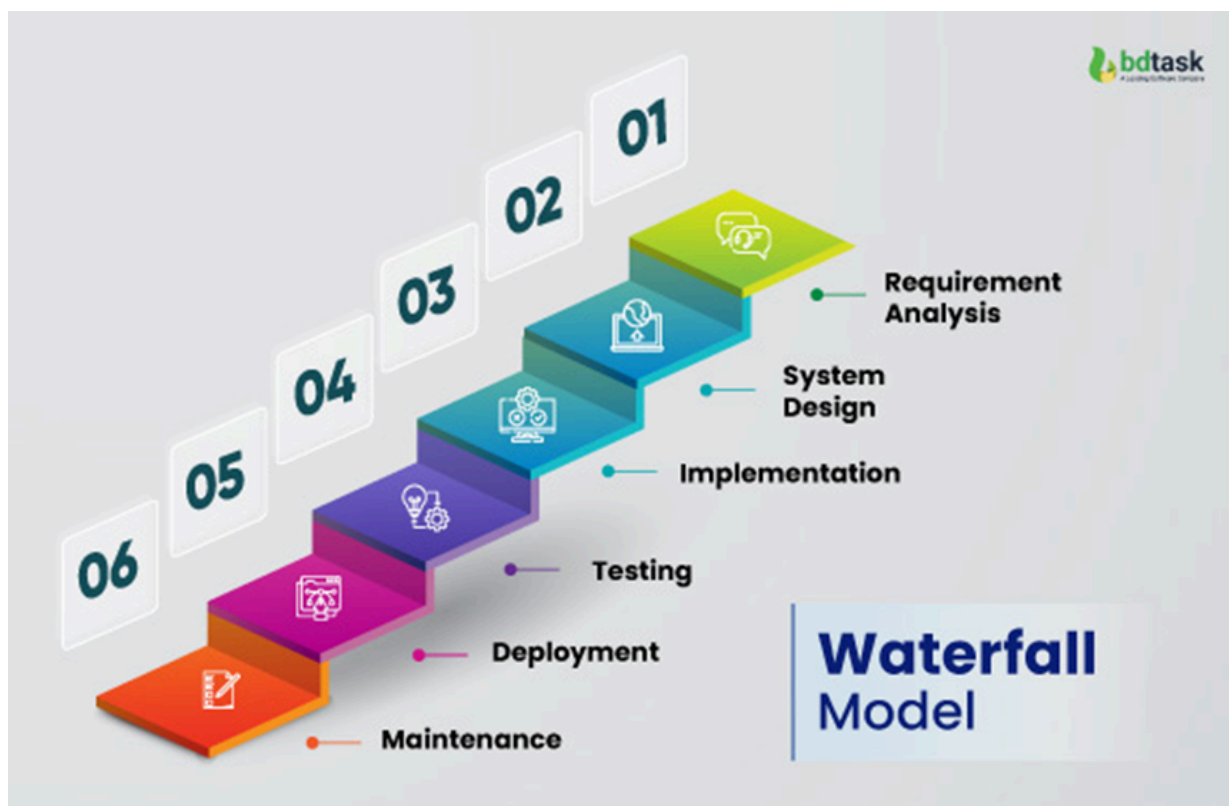
3.2 The Chosen Methodology

The Waterfall model under the System Development Life Cycle (SDLC) framework is selected as the development methodology for the PPG recoverable assets and inventory risk management system. The structure of this model is illustrated in Figure 3.1, which presents a linear sequence of development phases from requirement analysis through to maintenance (Simran, 2025).

The Waterfall model is suitable for this case study because the project scope, data sources, and analytical objectives are clearly defined at the initial stage. The inventory datasets, including materials, inventory, production orders, purchase orders, suppliers, and sales data, are already identified. Recoverable Assets (RA) and Magna Carta (MC) risk analysis as the expected outputs are also fixed. This reduces the need for iterative requirement changes and supports a structured development approach.

The system involves the integration of multiple data sources into a cloud-based environment using Microsoft Azure and a Medallion Architecture. As the data pipeline and transformation logic must be clearly defined before implementation, a sequential methodology ensures proper planning and reduces the risk of design inconsistencies (Simran, 2025).

Figure 3.1 reflects the disciplined flow required for this project, where each phase must be completed and validated before proceeding to the next (Simran, 2025). This is particularly important for ensuring the accuracy and reliability of financial and inventory risk outputs generated through the Power BI dashboard.



masukkan fasa yang dibuat dlm projek ini shj

Figure 3.1 Waterfall Model (Simran, 2025)

3.3 Phases of the Chosen Methodology

The Waterfall System Development Life Cycle is divided into several phases. Each phase produces a clear output serve as input for next phase. These approaches ensured that the requirements are well-defined and the project's goals are clear.

Phase 1: Requirement Analysis

[how to get the requirement???](#)

[what is other activities in requirement analysis us in this project](#)

This phase focuses on analyzing the current data environment within the PPG inventory management context to understand how raw materials, inventory, production, and sales data are currently managed. The study examines multiple disconnected datasets, including materials, inventory snapshots, purchase orders, production orders, suppliers, and sales orders, to identify how data flows across supply chain operations. Through this analysis, several key issues are identified, such as data silos, lack of integration, delayed reporting, and reliance on manual analysis for detecting inventory risks. These challenges make it difficult to identify Recoverable Assets (RA) and Magna Carta (MC) risks in a timely manner. Based on these findings, high-level system requirements and project boundaries are defined to ensure the proposed data engineering solution addresses end-to-end inventory visibility and risk detection rather than isolated data processing tasks.

Phase 2: System Design

[what is the activity in the system design use in this project](#)

In this phase, the overall architecture of the inventory risk analytics system is designed based on the requirements identified earlier. The system adopts a cloud-based Medallion Architecture consisting of Bronze, Silver, and Gold layers to structure data processing. The Bronze layer handles raw data ingestion, the Silver layer focuses on data cleaning and integration, and the Gold layer supports analytical modeling using a Star Schema. The relationships between datasets such as inventory, production, and sales are defined to ensure seamless data flow across the supply chain. Additionally, data models and pipeline workflows are designed to support the computation of RA and MC risks, financial provisioning, and inventory performance metrics. This phase ensures that the system architecture is scalable, consistent, and aligned with modern data engineering practices before moving to implementation.

Phase 3: Implementation

During the implementation phase, the designed architecture is developed into a functional data pipeline on the Microsoft Azure platform. Azure Blob Storage is used to store raw and processed data, while Azure Data Factory is utilized to orchestrate data ingestion and transformation processes following the ELT approach. The six datasets are ingested into the Bronze layer, transformed and cleaned in the Silver layer, and structured into a Star Schema in the Gold layer. Business logic for detecting RA and MC risks is implemented at this stage. In addition, Microsoft Power BI is used to develop interactive dashboards that visualize inventory health, financial impact, and supply chain risks. The implementation strictly follows the system design to ensure consistency and reduce development errors.

Phase 4: Testing

This phase involves validating the functionality and accuracy of the developed data pipeline and analytics system. Data validation testing ensures that transformations are correctly applied and that the outputs from the Silver and Gold layers are accurate and consistent. Functional testing is conducted to verify that RA and MC risk calculations, financial provisioning, and inventory metrics are correctly generated. Integration testing ensures that data flows smoothly across different layers of the Medallion Architecture and between Azure services and Power BI dashboards. Any identified issues, such as incorrect calculations, missing data, or inconsistencies, are documented and resolved to improve the reliability and performance of the system.

Phase 5: Deployment [ini buat k?](#)

In this phase, the completed system is prepared for deployment within a simulated production environment on Microsoft Azure. The data pipeline is configured to enable automated data ingestion, transformation, and reporting processes. Power BI dashboards are published and made accessible to stakeholders for monitoring inventory risks in real time. This phase includes configuring access permissions, ensuring proper integration between Azure services, and validating that the entire workflow operates as intended. Although this project is implemented as a case study, the deployment phase demonstrates how the proposed solution can function in a real-world enterprise environment to support data-driven decision-making.

Phase 6: Maintenance ini buat k

The maintenance phase focuses on ensuring the long-term performance and reliability of the inventory risk analytics system. This includes monitoring data pipeline performance, fixing errors, and updating transformation logic when business rules or data structures change. Maintenance activities also involve improving dashboard visualizations and incorporating user feedback to enhance usability and analytical capabilities. Although full-scale maintenance is beyond the scope of this project, this phase highlights the importance of continuous system support and scalability to adapt to evolving supply chain requirements and increasing data volume

3.4 Project Planning Schedule

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3.4.1 Gantt Chart

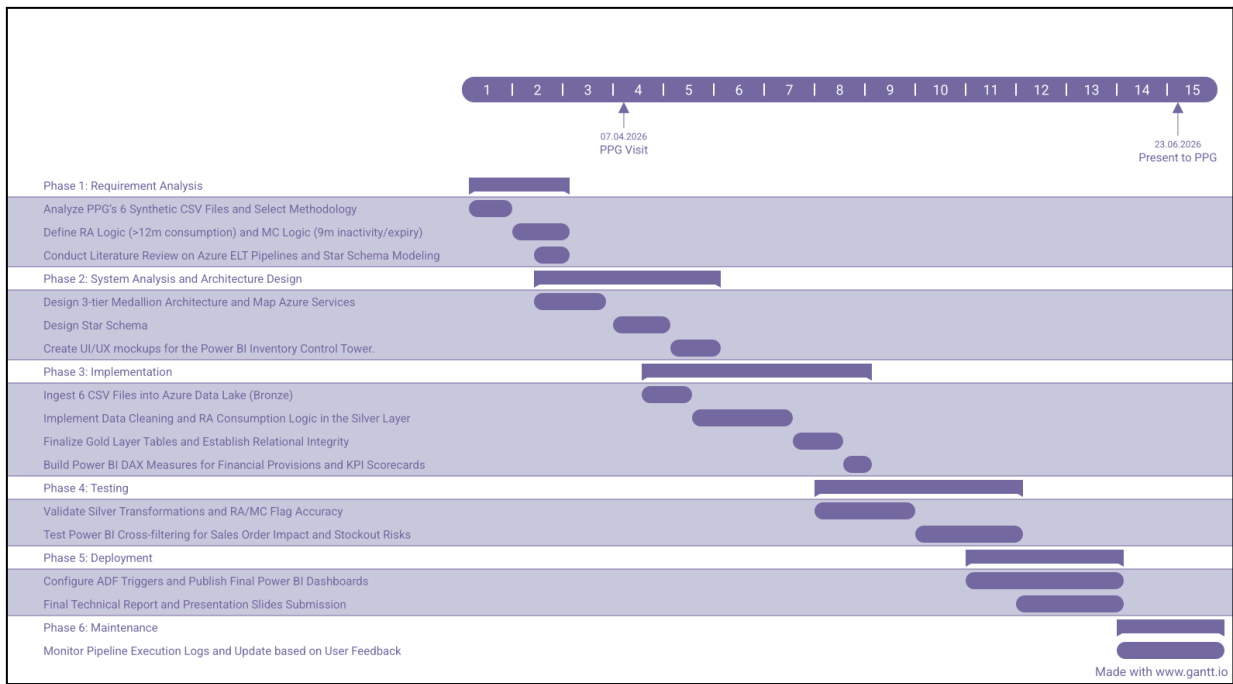


Figure 3.2 Project Schedule redraw

The project was initiated on March 16, 2026, with Phase 1: Requirement Analysis that aimed to determine data silos in the context of PPG inventory and establish the fundamental business logic of RA and MC risks. This stage was completed by March 30, giving the required background of Phase 2: System Analysis and Architecture Design. At this design stage, the group sketched the three-tier Medallion Architecture (Bronze, Silver, and Gold) and developed a central Star Schema to enable a robust reporting with analytics. The important milestone was a physical PPG Industrial Visit on April 7, 2026, to align the system design with the real-world operational requirements.

The project is designed to shift to Phase 3: Implementation at the end of April when the team will use Microsoft Azure services, such as Azure Data Factory to coordinate the processes and Azure Data Lake to ingest raw data (Bronze layer). The development milestones, including coding DAX measures in Power BI and implementing RA/MC flags in the Silver layer, will be completed to meet the draft deadline of May 11, 2026.

Then, Phase 4: Testing will be conducted to confirm the correctness of financial provisioning and data pipeline integrity as a whole. The project will proceed to Phase 5: Deployment which involves setting up automated triggers and completing the technical report before June 15, 2026. The final deliverable will be on June 23, 2026, with the official transfer of the complete analytics solution to PPG stakeholders. Lastly, Phase 6: Maintenance will develop a system of constant monitoring of the pipeline and its periodic update in accordance with the feedback of stakeholders.

3.5 Summary

This chapter has described the methodological approach that was taken to design the PPG inventory risk management system based on the SDLC framework. In particular, to guide the project, the linear Waterfall model was selected. The main outcome of the choice of this particular methodology is a highly organized development roadmap that is important since the parameters of the project, such as the six raw operational datasets, or the desired outputs of RA and MC risks, were clearly defined in the first place. The methodology is a systematic sequence of six controlled steps which include Requirement Analysis, System Design, Implementation, Testing, Deployment, and Maintenance. Through setting clear deliverables at every level, the

product of one phase is the input to the next. Indicatively, the existing constraints of disjointed data silos were analyzed to directly shape the next phase of creating a three-tier Medallion Architecture on Microsoft Azure. The strategic benefit and subsequent success of this Waterfall method are that it is phase-gated. Since the data pipeline architecture and ELT transformation logic need to be fully defined prior to any implementation, the methodology proactively mitigates the risk of design inconsistencies. Moreover, the single Testing step will make sure that the intricate logic of the financial provisioning of the RA and MC is strictly tested before the ultimate Power BI dashboards are launched. Finally, such a rigorous methodological approach will guarantee that the analytics system provided are not only reliable and scalable but also appropriately address the fundamental goal of moving the organization to a proactive and data-driven inventory approach.

CHAPTER 4

ANALYSIS AND DESIGN

4.1 Introduction

In Chapter 4, the comprehensive analysis and system design blueprints for the proposed automated Recoverable Assets (RA) and Inventory Risk Management (IRM) platform at PPG Industries Malaysia were presented. This chapter brings the high-level project lifecycle phases described in the project methodology down to the granular level, detailing the logic, functional boundaries, and architecture that is needed to build an end-to-end cloud data pipeline with Microsoft Azure and Power BI. The main goal is to get rid of the plant's current manual auditing processes and to implement an automatic system that will process non-uniform operational data to dynamically identify inventory obsolescence and measure the financial provision liabilities.

4.2 System Analysis

[explain](#)

4.2.1 Case Study

PPG Industries Malaysia has a well-equipped chemical plant that makes high-tech industrial coatings, car finishes, and building coatings. In this fast-paced operational environment, the control of raw material cycles is crucial to the liquidity of the supply chain, to the reduction of overheads and to the correct valuation in the balance sheet. One of the main operational threats faced by the facility is inventory obsolescence, which occurs when the amount of physical stock in the warehouse continuously exceeds the amount of goods produced on the spot.

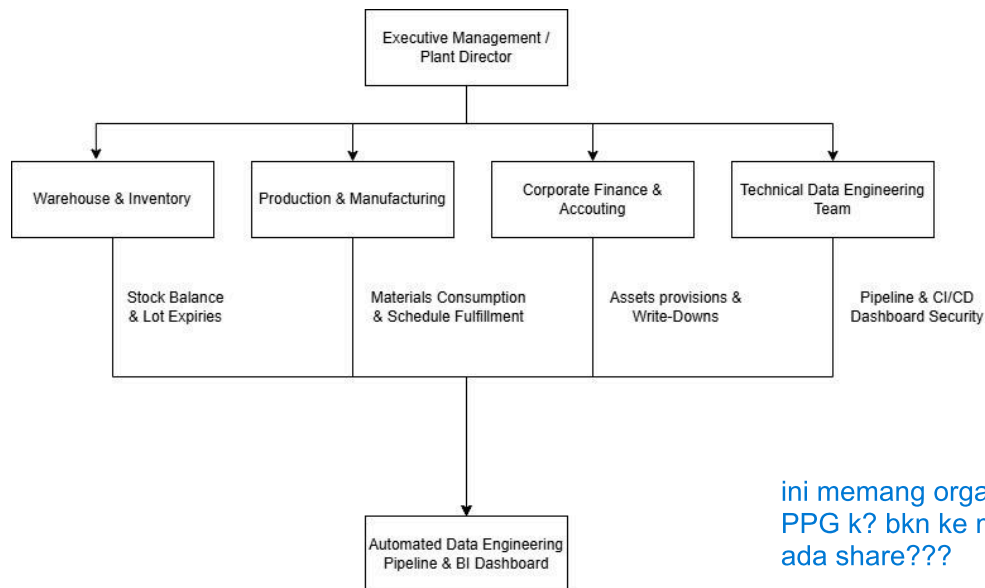
PPG uses two different internal financial and operational risk classification to quantify and manage material risks of dead, expiring or excess inventory.

- **Recoverable Assets (RA):** Raw materials that have a higher number of physical units on hand (based on the inventory_snapshot.csv) than the expected 12-month moving manufacturing consumption (based on the production_orders.csv). These materials are tied up working capital and should be reviewed quarterly, in a formal manner, to evaluate recovery options or reallocation to other production.
- **Magna Carta (MC):** The most severe material obsolescence is a critical sub-classification of RA. The category is divided into two sub categories: MC Dead Stock (materials with no manufacturing consumption for 9 months or more) and MC Expired (materials with the certified lot expiry date passed). Magna Carta inventory is a total asset loss in accordance with strict corporate compliance rules and requires a strict 100% financial write down provision today.

The data ecosystem that is the context of this problem space has six separate source files: materials.csv (structural material dimensions, safety margins, base costs), suppliers.csv (vendor metadata and reliability ratings), inventory_snapshot.csv (monthly localized warehouse stock balances across locations WH-A, WH-B, WH-C, WH-D, and WH-E), purchase_orders.csv (inbound supply chain and lead-time data), production_orders.csv (historical manufacturing material consumption), sales_orders.csv (outbound customer commitments). System analysis is used to determine how these isolated transactional components can be linked together in an automated cloud analytics system to overcome financial volatility and operational delays.

4.2.2 Company Organization Structure

The automated RA and Inventory Risk Management data pipeline creates a critical cross functional link between four main divisions of PPG Malaysia, namely, Warehouse Management, Production & Manufacturing, Corporate Finance & Accounting and the Technical Data Engineering Team.



ini memang organization structure dlm PPG k? bkn ke masa industry visit PPG ada share???

Figure 4.1: Functional Organizational Structure of PPG Industries Malaysia demonstrating cross-functional data dependencies and integration points for the automated Inventory Risk Management (IRM) data pipeline.

- Warehouse and Inventory Management:** This unit oversees the physical storage and handling of raw materials in decentralised plant sites (WH-A, WH-B and WH-C). They provide the basic inventory logs (`inventory_snapshot.csv`). To monitor material shelf-lives, they employ the central pipeline, and physically separate expired batches from active batches in the racks prior to production line distribution.
- Production and Manufacturing Team:** This unit performs the physical coating formulations and uses raw materials as per the active schedules. They create the metrics of consumption that are recorded in the `production_orders.csv`. The production team cross-references the predictive stockout warnings on the dashboard with the raw material's historical 12-month moving averages to avoid any factory downtime due to unfulfilled material requirements, which leads to expensive production losses.
- Corporate Finance and Accounting Division:** This team is responsible for fiduciary aspects of plant asset tracking and balance sheet integrity. They use the automated data pipeline extensively to make dynamic financial provisions. The finance team can create auditable, compliant 100% financial write-down reports without manually manipulating spreadsheets, as it automatically pulls out Magna Carta dead and expired stock and multiplies by verified base asset costs from `materials.csv`.

- **Technical Data Engineering Team:** This team is the foundation of the automated infrastructure. They are tasked with ensuring pipeline orchestration in Azure Data Factory, data quality rules in the Medallion architecture, database schema integrity for the analytical Star Schema data warehouse, and row-level security permissions for user access on the Power BI dashboard interface.

4.2.3 System Requirements Gathering Techniques

Instead of using traditional user interviews, this project's requirements were gathered using an analytical and organized method designed for data engineering pipelines and case study frameworks. The techniques are:

- **Document Analysis and Business Logic Synthesis:** Analysis of enterprise inventory tracking documents was carried out to identify the operational boundaries of inventory risks. To be more specific, risks of Recoverable Assets (RA) and Magna Carta (MC) were identified and taken out. Then, it will be converted into strict technical data transformation rules.
- **Dataset Schema Inspection:** Throughout all six core CSV files that were given, this team performed investigation and review on each file. It is to find the details for data ingestion and relatable requirements. So, this allows the team to identify such as primary keys, foreign keys etc. that are required to design the data pipeline.
- **Domain Benchmarking:** This team also gathers technical requirements for cloud architecture. It is by analyzing industry-standard frameworks and documented case studies of global manufacturing enterprises such as Epiroc and WNS. From there, it give the team more vision on technical specifications required to model the three-tier Medallion Architecture. Not to forget, this team can select the appropriate cloud stack.

4.3 System Requirements

[explain](#)

4.3.1 Functional Requirements

- The system must automatically ingest the six core CSV datasets from their source paths into the Bronze layer of Azure Data Lake Storage using Azure Data Factory orchestration.
- The system must utilize Azure Databricks to clean raw data.
- The system needs to calculate a rolling 12-month consumption forecast for each raw material to evaluate stock levels.
- The system needs to flag any raw material stock quantities exceeding the 12-month as a Recoverable Asset.
- The system must identify materials with zero operational consumption for 9 consecutive months or those past their expiration dates.
- The system must automatically map material shortages to specific customer sales orders.
- The system must deliver an interactive Power BI dashboard.

4.3.2 Non-Functional Requirements

- The system should automatically adjust its computing power and storage size based on how much ERP data is being processed, without needing people to manually upgrade servers.
- The system should organize data in a way that makes reports and dashboards run very fast, especially in Microsoft Power BI.
- The system must keep an original copy of all source data exactly as it was received.
- The system should continue from the failed step instead of restarting everything from the beginning if part of the pipeline fails.
- The Microsoft Power BI dashboard should feel smooth and responsive when users interact with it.

4.4 Current System Analysis

[explain](#)

4.4.1 Legacy Workflow Description

The current legacy operational system used for the PPG Industries Malaysia plant for inventory risk assessment and asset write-down provisions is completely manual, disjointed and spreadsheet-based. This is done on a non-real time basis at the end of each fiscal quarter. The workflow starts by inventory officers manually downloading the separate transactional data logs from the isolated localized databases and saving the individual logs as localized desktop spreadsheets. The operators are then forced to try to join these disjointed files with local spreadsheet software, taxing local computing resources. An analyst would have to create complicated nested array formulas to cross-reference the physical stock balance with historical manufacturing logs to calculate the 12-month moving average consumption for any given raw material. This sequence has to be followed line by line, across thousands of different material batches, causing a significant amount of operational friction and labour hours that can be used for better supply chain optimization.

4.4.2 Current Data Flow and Inventory Evaluation Logic

The manual system tries to assess risks to inventory based on information from various disjointed files, but there is no integrated database structure or formal entity relationships. Stock quantities are read from the inventory snapshot file, and the unit cost file should be read from the master materials sheet. Recoverable Assets are identified by manually calculating total consumption from historical production orders in the previous 12-months and then marking the material as an excess asset if there is more on hand than this computed amount. To determine Magna Carta Dead Stock the operator must perform a row-by-row visual inspection of production logs to find material identifiers that have zero usage flags for nine months or more. Finally, Magna Carta Expired stock is determined by performing a static inventory check for whether or not a batch has passed its certified lot expiry date. After these groups are manually separated, the financial provision is then calculated on another sheet, using obsolete quantities multiplied by the costs of the base unit of the file, which is very dependent on the structural integrity of the unmanaged files.

4.4.3 Identification of Operational Bottlenecks and Vulnerabilities

An in-depth examination of this manual architecture uncovers significant systemic weaknesses that impact supply chain performance and financial reporting precision. Firstly, the architecture is plagued by data latency and data siloing as data streams are derived from procurement, warehouse tracking and manufacturing logging systems which are independent and unconnected. These logs can be consolidated once every few weeks and the reports are static and retroactive, so the warehouse team never knows which lots are expiring or when their material needs are going to go out of stock and cause a disruption in their active production schedule. Moreover, the human data entry and manual calculation of spreadsheets mean that there is a high likelihood of error, with small syntax errors in the nested formulas, external workbook links or accidental row deletions skewing the final calculations and potentially impacting the integrity of financial provisions reporting to corporate stakeholders. Lastly, the legacy system doesn't provide a trace of downstream effects, and the warehouse team cannot relate a shortage in raw materials to a customer commitment that is on the go. If a material falls below the safety level, there is nothing in place to cross reference material lead times to open sales orders and predict which ones will be at risk of being delayed, which then impacts on client relationships and plant reliability.

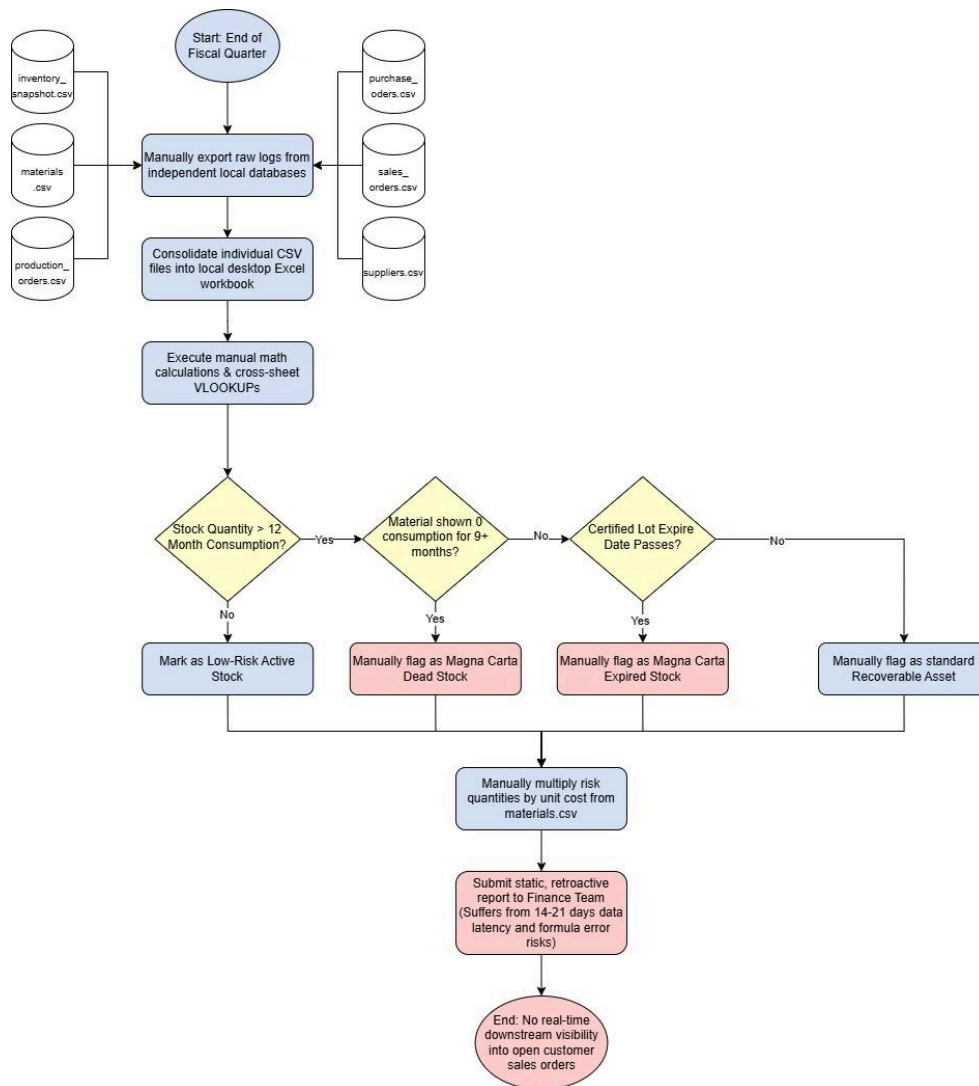


Figure 4.2: Flowchart of the legacy manual inventory evaluation process highlighting conditional risk verification checkpoints, severe data latency gaps, and human calculation vulnerabilities.

pastikan explain dlm teks

4.5 System Design

explain

4.5.1 System Architecture

buat satu lagi conceptual data architecture?/

Figure 4.3 below illustrates the cloud system architecture of the PPG data engineering pipeline. The diagram shows raw operational data being ingested from a local machine through Azure Data Factory and stored in a Data Lake. The data is then processed and transformed across three progressive layers using Azure Databricks. Once refined, it is loaded into Azure Synapse Analytics to enable centralised data storage and analytical access, before being visualised through Power BI.

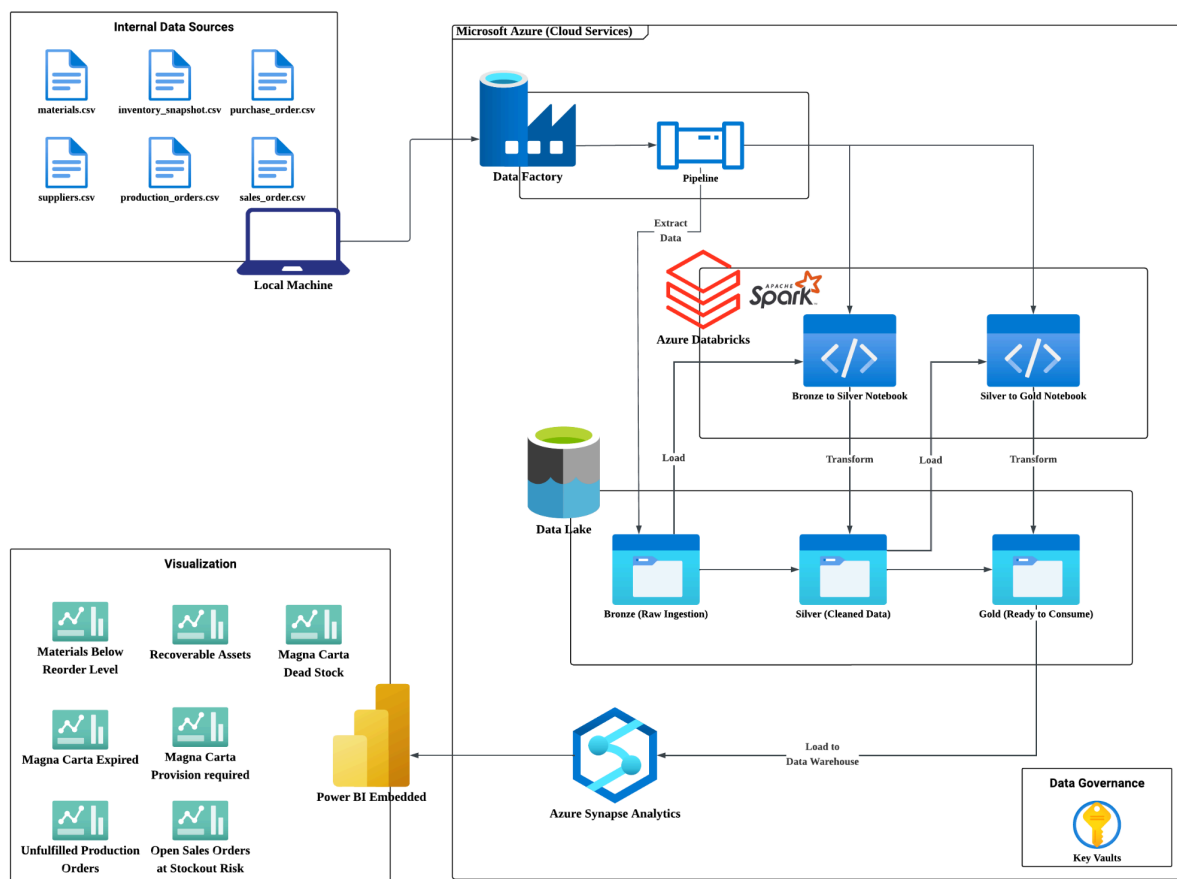


Figure 4.3: System Architecture

physical

4.5.2 System Architecture Component

This section describes the system architecture of the PPG data engineering pipeline. As illustrated in Figure 4.3, the system is built on a cloud-based infrastructure hosted on Microsoft Azure. It follows the Medallion Architecture pattern that progressively refines raw data through three quality layers before it is presented to end users as interactive dashboards.

The pipeline begins with six CSV files stored on a local machine, which include `materials.csv`, `inventory_snapshot.csv`, `purchase_order.csv`, `suppliers.csv`, `production_orders.csv`, and `sales_order.csv`. These files represent the core operational data of the organisation and serve as the primary data source for the entire pipeline. To bring these files into the cloud, Azure Data Factory (ADF) is used as the orchestration tool. It is responsible for extracting the raw CSV files from the local machine and triggering the pipeline that governs the flow of data into the Azure environment.

Once the data is ingested, it is processed within Azure Databricks, which is powered by Apache Spark to handle large-scale data transformation. The transformation is carried out through two notebooks, namely the Bronze to Silver Notebook, which cleans and validates the raw data, and the Silver to Gold Notebook, which applies business logic to produce analysis-ready data. Throughout this process, data is stored in a Data Lake that is organised into three layers. The Bronze layer holds raw ingested data as-is, the Silver layer contains cleaned and validated data, and the Gold layer stores fully transformed data that is ready for business consumption. Each layer represents an increasing level of data quality to ensure that only reliable data reaches the next stage.

The refined data from the Gold layer is then loaded into Azure Synapse Analytics, which serves as the central data warehouse of the system. From there, Power BI Embedded connects to Synapse to retrieve the processed data and render it as interactive dashboards for business users. The dashboards cover seven reporting areas, which are Materials Below Reorder Level, Recoverable Assets, Magna Carta Dead Stock, Magna Carta Expired, Magna Carta Provision Required, Unfulfilled Production Orders, and Open Sales Orders at Stockout Risk. Finally, Azure Key Vaults is implemented across the entire architecture to enforce data governance by securely managing credentials and access keys. This is to ensure that only authorised services can interact with the pipeline.

4.5.3 Data Engineering Process

This data engineering process for the Recoverable Assets (RA) and Inventory Risk Management (IRM) system is designed using a modern Extract, Load, Transform (ELT) approach that leverages a three-tier Medallion Architecture on Microsoft Azure. Data Ingestion Strategy

The data pipeline starts with six synthetic datasets from different sources as its targets, which include `materials.csv`, `inventory_snapshot.csv`, `purchase_orders.csv`, `production_orders.csv`, `suppliers.csv`, and `sales_orders.csv`—the heart of the PPG supply chain. Controlling the flow of this data, Azure Data Factory is used as the main orchestration tool, with the ability to synchronize the ingestion and transformation processes. Bronze Layer (Ingestion & Cleaning Rules)

The Azure Data Factory copies the raw CSV files from the source to the Bronze layer (Azure Data Lake Storage Gen2 in this case). Auditable Raw Storage: Data is stored as is, without hard rules being applied to the data as it is entered. Pipeline Resilience: When you intentionally don't transform the data, you're able to keep a historic record of the data, and if the business logic later needs to be reconfigured, the entire pipeline can be reprocessed from this source of raw data. This is a silver layer (Transformation Business Logic).

After the data is staged in Azure Databricks, the complex business logic is applied to the data and joins it together and cleans it to create the silver layer. To enforce PPG's strict inventory logic, Databricks efficiently performs window operations and date comparisons on the very large sets of data. The transformation rules applied are: The following transformation rules will be applied:

Rolling Consumption Calculation: The system will calculate the actual 12-month rolling manufacturing consumption for each material based on historical data provided in the `production_orders.csv` file.

Recoverable Assets (RA) Identification: It is the pipeline that compares the forecast of consumption for the last 12 months with the physical units that are on hand, and these physical units are taken from `inventory_snapshot.csv`. If the inventory level exceeds the forecast level for the next year, it is classified and marked as an RA (Excess Stock).

Magna Carta (MC) Risk Detection: The system performs a more extensive obsolescence classification, which determines a serious loss of assets, and is broken down into two sub-classifications:

MC Dead Stock: Materials with no manufacturing use whatsoever for 9 months or more.

MC Expired: Materials that have a certified lot expiration date that has passed.

Auto-Financial Provisioning: If the material is flagged as MC, then the standard accounting principle is that it is no longer an asset. The Silver layer logic dynamically computes a 100% financial write-down, which is the number of obsolete physical quantities multiplied by the base cost of the material in materials.csv.

4.5.4 Database Design

mana ERD? bkn dlm arahan PPG mintak buat ERD

To effectively manage and analyze the complex inventory data at PPG, the analytical data model is structured using a Fact Constellation Schema which is an advanced Star Schema. This dimensional modeling approach minimizes complex joins, allowing visualization tools like Microsoft Power BI to aggregate millions of rows efficiently and deliver real-time insights into inventory health.

Rather than forcing all operational data into a single table, this schema centers on a primary fact table, fact_inventory_status, which stores the quantitative stock metrics and computed risk indicators including RA and MC flags. This central table operates alongside secondary fact tables such as fact_purchase_orders and fact_sales_orders that share conformed dimension tables. This interconnected structure enables cross-functional filtering, such as instantly mapping a material shortage to the specific customer sales orders it will delay.

Data Dictionary and Schema Structure

Table Name	Table Type	Primary Key (PK)	Foreign Keys (FK) & Core Attributes

fact_inventory_status	Fact	inventory_status_id	FKs: material_id, warehouse_id, snapshot_date_id Metrics: batch_lot_number, physical_stock_quantity, rolling_12m_consumption, total_financial_provision, lot_expiry_date Flags: ra_flag, mc_dead_stock_flag, mc_expired_flag
fact_purchase_orders	Fact	po_id	FKs: material_id, supplier_id, order_date_id, delivery_date_id Metrics: order_quantity, received_quantity, po_status, lead_time_days
fact_sales_orders	Fact	sales_order_id	FKs: material_id, date_id, customer_id Metrics: order_quantity, fulfillment_status
dim_materials	Dimension	material_id	material_name, material_category, base_unit_cost, shelf_life_days, safety_stock_level

dim_suppliers	Dimension	supplier_id	supplier_name, region, vendor_rating, contact_info
dim_warehouse	Dimension	warehouse_id	location_name , capacity
dim_date	Dimension	date_id	full_date, year, quarter, month, day

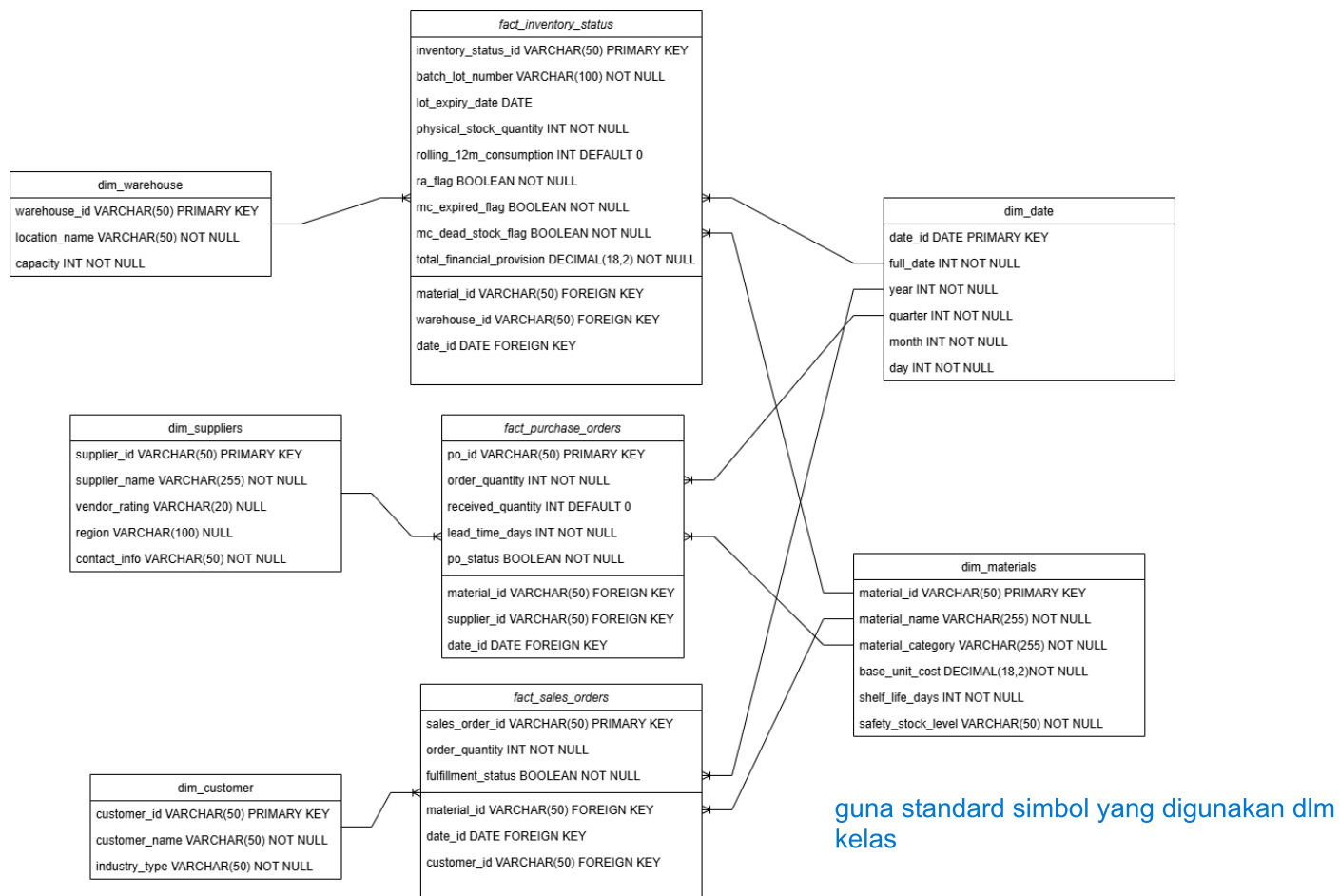


Figure 4.3: Database Design using Fact Constellation Schema

Table Relationships and Multiplicities

The data model enforces strict relational integrity through One-to-Many (1:N) relationships between the dimension tables and the fact tables. 1:N Material to Facts: A specific raw

material in `dim_materials` connects to multiple daily inventory snapshots, historical purchase orders, and outbound sales orders. This is the critical relationship that allows the Power BI dashboard to trace the downstream impact of an at-risk material. 1:N Supplier to Purchases: A single vendor in `dim_suppliers` connects to multiple historical and active inbound purchase orders in `fact_purchase_orders`. 1:N Warehouse to Inventory: One physical storage location in `dim_warehouse` links to thousands of daily material stock logs within that facility. 1:N Date to Facts: A single calendar date in `dim_date` connects to the inventory snapshots taken on that day, the purchase orders delivered on that day, and the sales orders required on that day.

Data Grain

The grain of the primary `fact_inventory_status` table is defined at the lowest possible atomic level: one row per material, per warehouse location, per batch/lot number, per daily snapshot. Maintaining the data at this high-resolution grain is an absolute necessity for computing the Magna Carta (MC) Expired risk. Because different batches of the identical raw material expire on different certified dates, summarizing data merely at the material level would obscure these lot-specific expiries. This granular structure ensures the Power BI dashboard can aggregate high-level financial write-downs for executive reporting while still allowing warehouse operators to drill down and locate specific expiring batches on the floor.

4.6 Summary

This chapter discussed the analysis and design of the RA and IRM system for PPG Industries Malaysia. In the analysis stage, it was considered necessary to look into the inventory management environment, organizational structures, system requirements, and shortcomings of the current manual method of inventory management. Several crucial problems have been found in the analysis stage; for example, information silos, delay in reporting, manual calculation, high probability of errors committed by the human operator, inability to detect obsolete inventory, and stock-out problems.

A cloud-based system architecture was developed based on the requirements analysis using Microsoft Azure technologies and the Medallion Architecture approach. The solution utilizes Azure Data Factory to manage the data flow, Azure Data Lake Storage to store the data in a

multilayered manner, Azure Databricks to transform the data and apply the business logic, Azure Synapse Analytics as a serving layer and Power BI for interactive data visualization. The data engineering was implemented with the usage of the ELT method allowing for automatic identification of RA, MC Dead Stock, MC Expired items and financial provisioning computation.

Moreover, the Fact Constellation Schema was introduced in order to facilitate effective data analysis and reporting. The dashboard design allows the company stakeholders to get access to the current status of the inventory health, potential stockouts, financial liabilities and customer orders at risk. Finally, the completed analysis and design activities provide a foundation for further implementation of the system to be considered in Chapter 6.

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 Introduction

This section details the practical execution and system implementation of the end-to-end cloud analytics pipeline for the PPG Recoverable Assets (RA) and Inventory Risk Management (IRM) project. Building upon the architectural blueprints and dimensional data models established in previous phases, this segment documents the physical deployment of data engineering processes within the Microsoft Azure ecosystem. It encompasses the complete lifecycle of the data from the automated ingestion of raw operational records into the data lakehouse, through the rigorous transformation stages of the Medallion Architecture, to the final high-performance visualization layer. The resulting infrastructure ensures a secure, scalable, and automated solution for mitigating inventory obsolescence risks.

5.2 Task Distribution

[masukkan dalm gantt chart](#)

Data Ingestion Bronze Layer

- **Assignee:** Lim Yu Han & Ng Yu Hin
- **Description:** Responsible for architecting the initial data extraction process. This involved configuring automated pipelines such as Azure Data Factory to securely ingest raw operational data files and land them in their immutable, original state into the data lake's Bronze layer.

Bronze to Silver Notebook

- **Assignee:** Ling Yu Qian
- **Description:** Tasked with the initial data cleansing and transformation phase using Databricks. This role required writing data processing notebooks to clean the raw data,

handle missing values, enforce correct data types, remove duplicates, and load the validated datasets into the Silver layer.

Silver to Gold Notebook

- **Assignees:** Nawwarah Auni Binti Nazrudin and Neo Li Xin
- **Description:** Entrusted with implementing the core business logic. This involved writing advanced transformations to calculate metrics such as the 12-month rolling consumption, generate Recoverable Asset (RA) and Magna Carta (MC) risk flags, and structure the refined data into a Star Schema before loading it into the Gold layer.

Synapse Analytics Setup

- **Assignee:** Lim Yu Han
- **Description:** Managed the data serving infrastructure. This task involved configuring Azure Synapse Analytics workspaces and serverless SQL pools to act as the optimized bridge between the structured Gold layer data and downstream business intelligence tools.

Power BI Dashboard (Data Visualization)

- **Assignee:** Ahmad Adib Zikri Bin A.Mazlam
- **Description:** Led the development of the front-end user experience. This involved connecting Microsoft Power BI to the serving layer to build an interactive, high-performance dashboard that translates the underlying data models into actionable visual insights for executives.

5.3 Development Environment Setup

The implementation of the PPG Inventory Risk Management pipeline relies on a robust and scalable cloud infrastructure. The development environment was provisioned entirely within Microsoft Azure, utilizing a single, centralized Resource Group (my-resource-group) to manage the lifecycle, integration, and security of all related architectural components.

Azure Services Provisioned

To construct the end-to-end Medallion Architecture, the following core Azure resources were deployed:

- **Azure Data Lake Storage Gen2 (ppgprojectdl):** Serves as the foundational enterprise data lake. It is configured with a hierarchical namespace to efficiently store and manage the raw, cleansed, and curated datasets across the Bronze, Silver, and Gold layers.
- **Azure Data Factory (ppgprojectdf):** Acts as the primary data orchestration and ingestion service. It is responsible for establishing automated pipeline connections to extract the operational source files such as inventory snapshots and purchase orders and load them into the Bronze layer.
- **Azure Databricks (dbw-ppgproject):** Deployed as the core computational engine. This collaborative, Apache Spark-based environment hosts the Python/PySpark notebooks used for executing complex data cleansing, schema enforcement, and the calculation of inventory risk metrics for the Silver and Gold layers.
- **Azure Key Vault (ppgprojectkv):** Implemented to ensure strict security governance. It securely stores sensitive credentials, secrets, and connection strings required for Data Factory and Databricks to authenticate and interact seamlessly without hardcoding passwords.
- **Azure Synapse Analytics:** Functions as the high-performance serving layer. It provides serverless SQL pools to query the highly structured Gold layer star schema, acting as the optimized bridge to the Power BI dashboard.

Resource Group Setup and Configuration

The visual evidence of the provisioned development environment within the Azure Portal is documented below.

pastikan setiap figure atau table ada penerangan dlm teks..pastikan lebbelling yang betul

Home > Resource Manager | Resource groups

ppg-project-rg
Resource group

How to manage these changes more efficiently with deployment tools? Export resource groups using Bicep or Terraform +1

Search

+ Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Group by none

Overview

Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Cost Management
Monitoring
Automation
Help

Essentials

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name	Type	Location
ppg-project-databricks	Azure Databricks Service	East Asia
ppg-project-df	Data factory (V2)	East Asia
ppg-project-kv	Key vault	East Asia
ppg-project-synapse	Synapse workspace	East Asia
ppgprojectsg	Storage account	East Asia

Figure 5.1: Overview of the primary Azure Resource Group, displaying the successfully deployed Data Lake, Data Factory, Key Vault, and Databricks service.

Home > Storage center | Blob Storage

ppgprojectsg
Storage account

Analyze connectivity to this storage account Show me alerts for this storage account Prevent unauthorized access to this storage account

Search

Upload Open in Explorer Delete Move Refresh Open in mobile CLI / PS Feedback

Overview

Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Partner solutions
Resource visualizer
Data storage
Security + networking
Data management
Settings
Monitoring
Automation

Essentials

Resource group (ppg-project-rg)
Location : eastasia
Subscription (move) : Azure for Students
Subscription ID : 75a4a525-3e70-4be8-839f-3e9a621bcbe3
Disk state : Available
Tags (edit) : Project : PPG-Analytics

Performance : Standard
Replication : Locally redundant storage (LRS)
Account kind : StorageV2 (general purpose v2)
Provisioning state : Succeeded
Created : 5/22/2026, 8:02:17 PM

Properties Monitoring Capabilities (5) Recommendations (0) Tutorials Tools + SDKs

Data Lake Storage

Hierarchical namespace	Enabled
Default access tier	Hot
Blob anonymous access	Enabled
Blob soft delete	Disabled
Container soft delete	Disabled
Versioning	Disabled
Change feed	Disabled

Security

Require secure transfer for REST API operations	Enabled
Storage account key access	Enabled
Minimum TLS version	Version 1.2
Infrastructure encryption	Disabled

Networking

Public network access	Enabled
-----------------------	---------

Microsoft Azure Upgrade

Search resources, services, and docs (G+/)

Copilot

ngghin@live.utm.my
UNIVERSITI TEKNOLOGI MALAYSIA

Home > ppgprojectsg

ppgprojectsg | Containers

Storage account

Search

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

Search containers by prefix Only show active containers

Showing all 3 items

Name	Last modified	Anonymous access level	Lease state	Storage connector
bronze	5/23/2026, 2:56:05 PM	Private	Available	-
gold	5/23/2026, 4:28:16 PM	Private	Available	-
silver	5/23/2026, 4:28:08 PM	Private	Available	-

Figure 5.2: Azure Storage account (ppgprojectdl) with Data Lake Storage Gen2 capabilities enabled for scalable cloud storage.

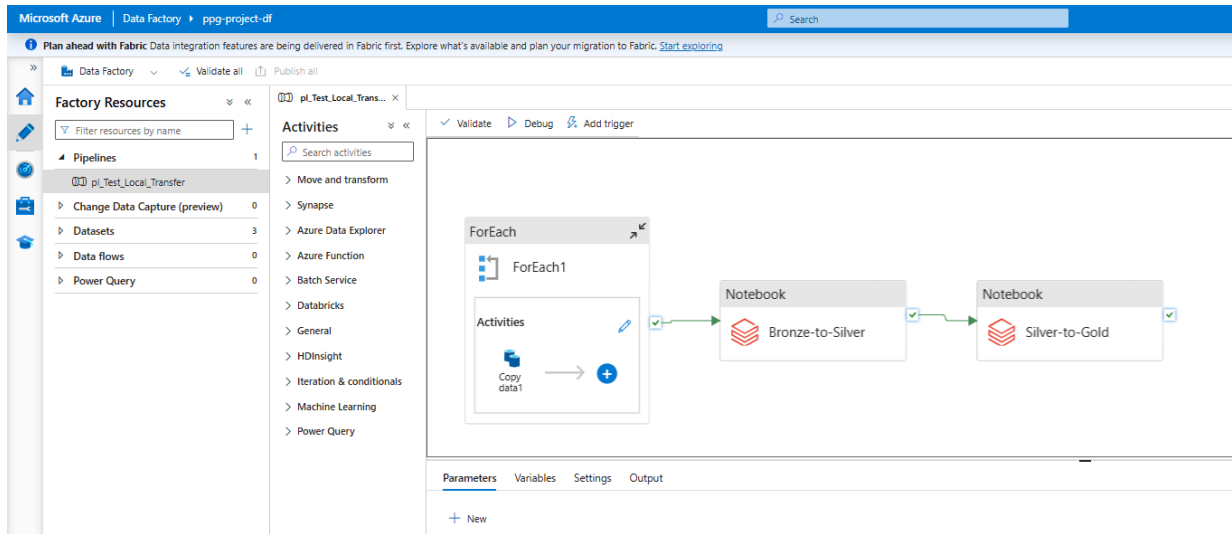


Figure 5.3: Azure Data Factory (ppgprojectdf) instance, provisioned for pipeline orchestration and data integration tasks.

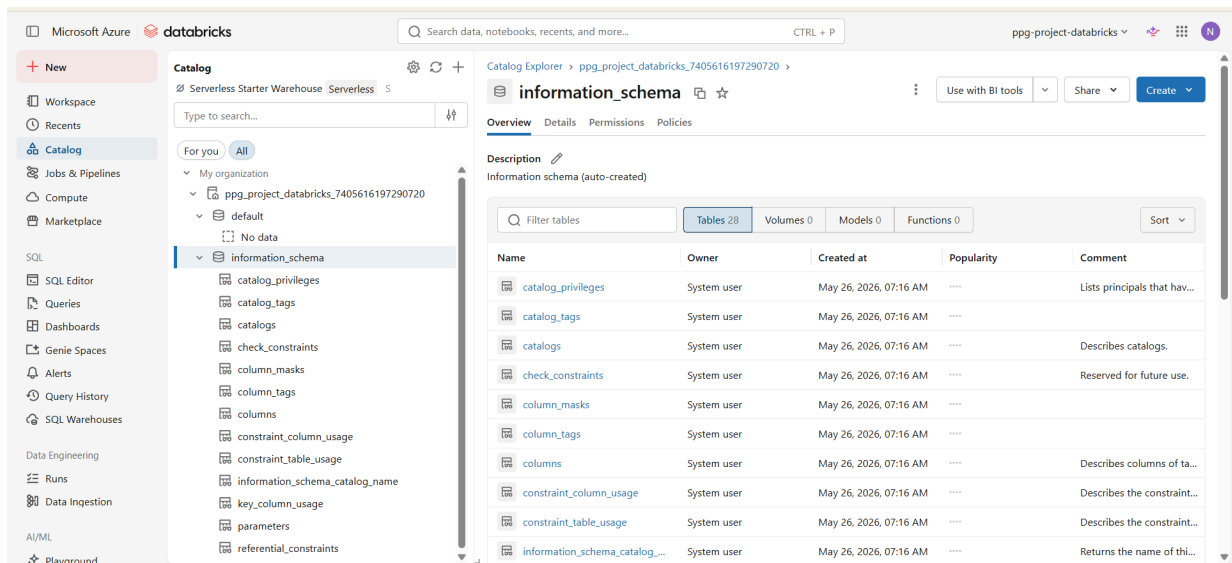


Figure 5.4: Azure Databricks workspace (dbw-ppgproject) for executing PySpark data transformations and inventory risk metric calculations.

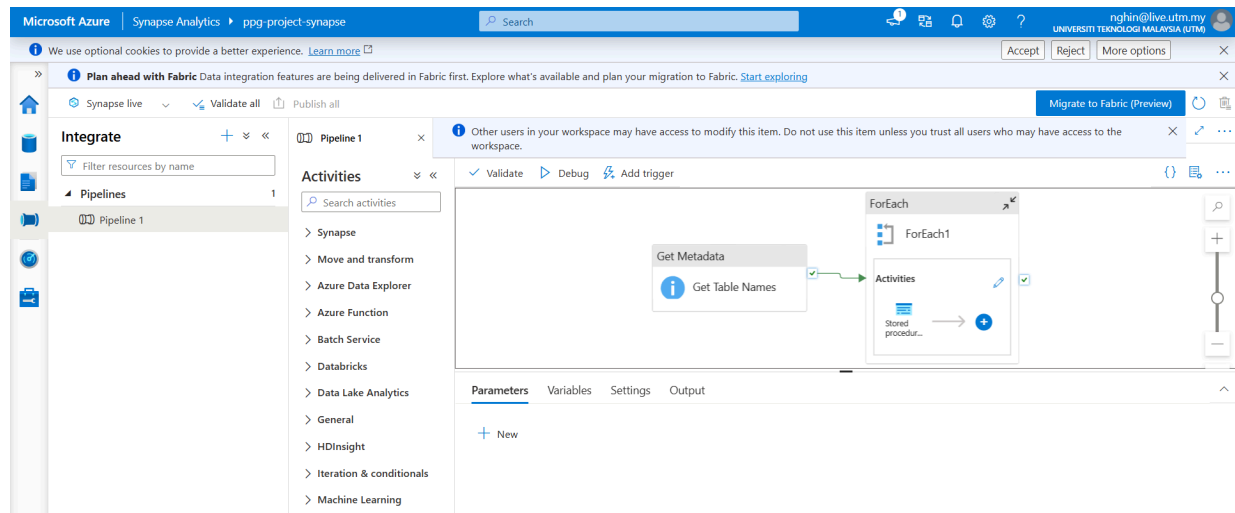


Figure 5.6: Azure Synapse Studio interface

5.4 Data Ingestion Bronze Layer

The initial phase of the system implementation focuses on the orchestration of data ingestion from the raw data repositories into the cloud environment. An Azure Data Factory (ADF) pipeline named 'pl_Test_Local_Transfer' was engineered to automate the migration process. To optimize ingestion efficiency, the pipeline utilizes a ForEach control activity paired with a dynamic Copy Data task. This structural choice enables the parallel processing of multiple data sources rather than forcing a slow sequential execution. The pipeline parameterizes the file paths, connecting to the source environment and safely streaming data into the cloud landing zone.

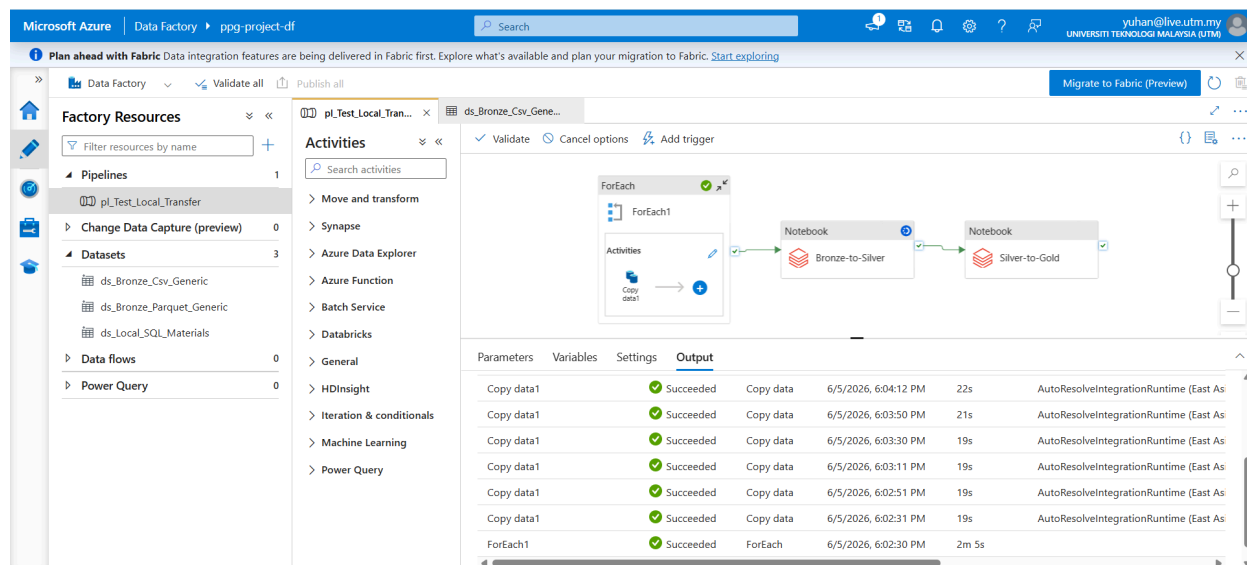


Figure 5.7: Azure Data Factory Orchestration Pipeline Execution

The destination for this raw ingestion is the designated Bronze container within Azure Data Lake Storage Gen2 (ADLS Gen2). The pipeline effectively moves all six required operational datasets namely materials.csv, suppliers.csv, inventory_snapshot.csv, purchase_orders.csv, production_orders.csv, and sales_orders.csv without changing their structural form. Storing data in its native format within the Bronze layer ensures a complete, historical audit trail is preserved for downstream processing.

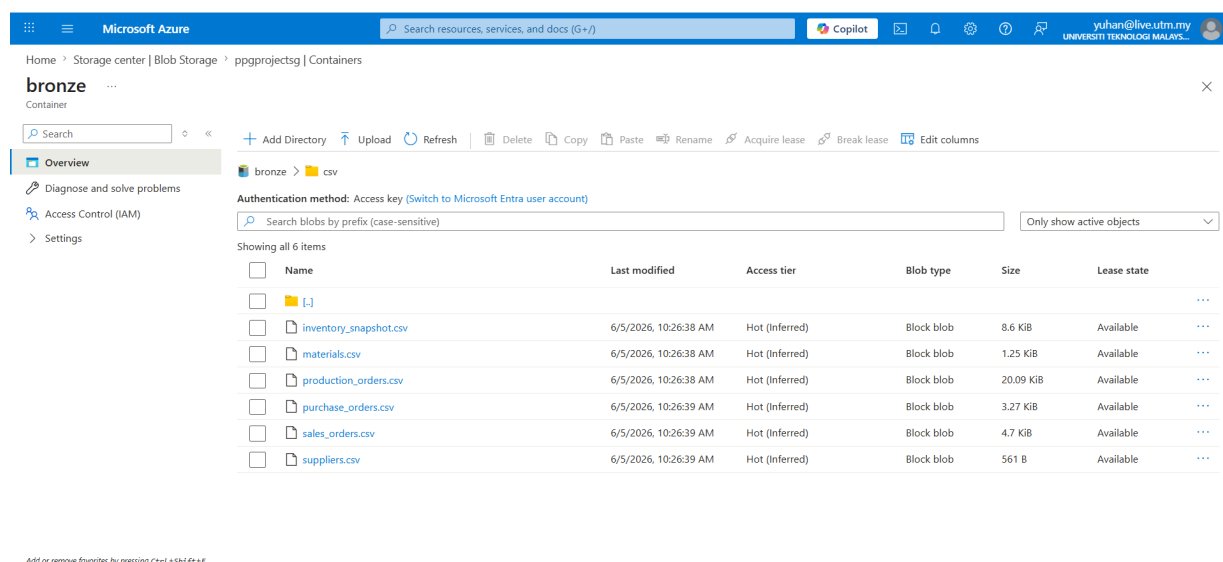


Figure 5.8: Azure Data Lake Storage Gen2 Container Structure

5.5 Data Transformation

[explain](#)

5.5.1 Bronze to Silver Notebook

This phase deals with data cleansing and transformation that starts out in Azure Databricks. The notebook is tasked with systematically cleaning the raw operational data and putting the cleaned data into the Silver layer.

- Data Cleaning Rules applied to the data.

The notebook uses a self-made function called `clean_staging_data` which goes through every data column in the notebook. The notebook includes a custom function called `clean_staging_data` which loops through all data columns. If it is determined to be a string type, then a `'trim()'` function is used on each column to remove any trailing or leading whitespaces from the text, thus ensuring it is formatted the same way.

Automated Iteration: An execution loop reads the raw `'parquet'` files in the Bronze container for each of the six core tables (`'materials'`, `'suppliers'`, `'inventory_snapshot'`, `'production_orders'`, `'purchase_orders'`, and `'sales_orders'`), and performs the cleaning rules uniformly.

- The handling of null values and duplicates.

Null Value Normalization: The script is designed to detect if there is any empty string value entered into the cell, and it will set the value of that cell to `'None'` (Null) to prevent any downstream calculation issues in financial provisions.

Absolute duplicate rows that could have been created while importing the raw data from legacy systems are properly eliminated with a `'dropDuplicates()'` command.

- Schema Enforcement

Data from the Bronze layer is explicitly rewritten from the `'parquet'` format into the very efficient `'delta'` format in Delta Lake Transition.

Strict Data Typing: The notebook requires that data be of a certain type because of `.option("overwriteSchema", "true")` to resolve schema mismatch and/or damaged raw data

instances. Upon overwriting of Silver tables, this command will force the tables to safely replace the old or corrupted schemas with the newly validated schemas.

```
Starting Bronze to Silver Processing Loop... 🚀
-----
✅ Table [materials] cleanly migrated to Silver.
✅ Table [suppliers] cleanly migrated to Silver.
✅ Table [inventory_snapshot] cleanly migrated to Silver.
✅ Table [production_orders] cleanly migrated to Silver.
✅ Table [purchase_orders] cleanly migrated to Silver.
✅ Table [sales_orders] cleanly migrated to Silver.
-----
All staging tables successfully standardized in Silver! 🎉
```

Figure 5.9: Bronze to Silver Processing Loop Execution Log

The Azure Databricks console execution log in Figure 5.9 shows that the Bronze to Silver data processing loop worked. As displayed in the output, the six core operational datasets (materials, suppliers, inventory_snapshot, production_orders, purchase_orders, and sales_orders) were processed one at a time without any systemic failures happening along the way. This console output provides technical evidence that the ELT (Extract, Load, Transform) pipeline, automated as outlined in the code above, successfully pulled the raw files from the Bronze landing zone, performed the cleansing transformations defined in the code provided above (such as null value detection and elimination, whitespace trimming, and duplicates), and ultimately loaded the validated records into the Silver layer. The fact that all tables were "successfully standardized" confirms the data is now structured, reliable and ready for complex business logic and metric calculations used in the next layer of notebooks in the Gold layer.

Silver Layer 'materials' Sample Validation:

Table	+						
material_id	material_name	category	unit_of_measure	reorder_level	lead_time_days	unit_cost	
1 MAT005	Chrome Yellow	Pigment	KG	100	21	6.400000095367432	
2 MAT001	Titanium Dioxide	Pigment	KG	500	14	8.5	
3 MAT002	Iron Oxide Red	Pigment	KG	300	10	3.200000047683716	
4 MAT007	Acrylic Resin	Resin	KG	600	14	5.300000190734863	
5 MAT019	Cardboard Box	Packaging	Unit	1000	7	0.44999998807907104	

Figure 5.10: Silver Layer 'materials' Sample Validation Output

The figure 5.10 is an example of material discovery for the materials dataset after it has been successfully loaded into the Silver layer of Azure Databricks. The following figure shows an example of material discovery for the materials dataset once it has been successfully loaded into the Silver layer of Azure Databricks. This is a visual example showing that the raw data has been cleaned, parsed and converted to a clean tabular form with enforced schemas.

Key attributes of the material are displayed as output, which are inputs to further calculations downstream for Recoverable Assets (RA) and Magna Carta (MC) risk. For example, the `unit_cost` column is correctly formatted as a floating-point number, a requirement for the 100% provisions for write-down of expired stock, which is required by the Finance team. Also, the operational measures like `reorder_level`, `lead_time_days` are very well defined and easily available for loading into a Star Schema in the Gold layer for the final Power BI Dashboard integrations. The consistent formatting of the text in all the categorical columns also confirms that the whitespace trimming and null handling logic in the notebook is well executed.

```
✓ [materials] schema force-reset. Columns: ['material_id', 'material_name', 'category', 'unit_of_measure', 'reorder_level', 'lead_time_days', 'unit_cost']
✓ [suppliers] schema force-reset. Columns: ['supplier_id', 'supplier_name', 'country', 'reliability_rating']
✓ [inventory_snapshot] schema force-reset. Columns: ['snapshot_date', 'material_id', 'warehouse_location', 'quantity_on_hand', 'expiry_date']
✓ [production_orders] schema force-reset. Columns: ['production_order_id', 'production_date', 'finished_product', 'material_id', 'quantity_consumed']
✓ [purchase_orders] schema force-reset. Columns: ['po_id', 'po_date', 'supplier_id', 'material_id', 'quantity_ordered', 'quantity_received', 'expected_delivery_date', 'actual_delivery_date']
✓ [sales_orders] schema force-reset. Columns: ['sales_order_id', 'order_date', 'customer_name', 'finished_product', 'quantity_ordered', 'required_delivery_date', 'status']
```

Figure 5.11: Schema Force-Reset Execution Log for Silver Layer Tables

Figure 5.11 shows that the schema enforcement logic was successfully executed when it was run in the Azure Databricks console. The "schema force-reset" messages refer to the script using the `overwriteSchema` command to replace any data structure that failed to match or was corrupted in the Silver layer. The output clearly specifies the exact column mapping used for all six core datasets (materials, suppliers, inventory_snapshot, production_orders, purchase_orders, and sales_orders), with important data attributes like `quantity_on_hand`, `unit_cost`, and `material_id` being properly organized. This stringent structural validation ensures the integrity of the data at the time they move out of the Bronze layer such that the datasets are exactly structured for the higher-level Recoverable Assets (RA) and Magna Carta (MC) metric calculations in the following Gold layer.

```
[ 'snapshot_date', 'material_id', 'warehouse_location', 'quantity_on_hand', 'expiry_date' ]
```

snapshot_date	material_id	warehouse_location	quantity_on_hand	expiry_date
2025-01-31	MAT001	WH-A	2000	2025-12-31
2025-02-28	MAT001	WH-A	2960	2025-12-31
2025-03-31	MAT001	WH-A	1810	2025-12-31

only showing top 3 rows

Figure 5.12: Data Validation Output for the inventory_snapshot Table

The data validation check for the inventory_snapshot data can be seen in Figure 5.12 below for a typical PySpark dataframe display command. The output of the console confirms the precise structure of the columns that were parsed by the script—snapshot_date, material_id, warehouse_location, quantity_on_hand, and expiry_date. The engineering team can then output top three rows of the dataset to see if the physical stock balances from localised warehouses (e.g. location WH-A) are correctly mapped and formatted. The integrity of these specific data points is important, as the total on hand and expiry date are the most fundamental data points to help determine Recoverable Assets (RA) and Magna Carta (MC) obsolescence risks in the future.

5.5.2 Silver to Gold Notebook

With the Silver layer established, the next stage involved transforming the cleaned and validated data into business-ready Gold tables using Apache Spark on Azure Databricks. The Gold layer serves as the final analytical layer. It contained KPI metrics derived from multiple Silver tables.

Before the transformation logic could be applied, the necessary libraries were imported and the Azure Data Lake Storage connection was configured. The storage account ppgprojectsg was referenced using an account key, and a helper function read_silver() was defined to streamline the reading of Delta tables from the Silver container, as shown in Figure 5.13.

```

# Setup & Load Silver Data
from pyspark.sql.functions import col, sum, avg, when, current_date, datediff, to_date

# Configuration
STORAGE_KEY = "9v/QDzffL1o0unVZpaLxKGPfCvTVkML3ZP1BJA67IRF4uBmQJ1ajvwb8aY4SGtrh98fbU0XYEVHP+ASTK8KBjQ=="
STORAGE_ACCOUNT = "ppgprojectsg"
BASE_URL = f"abfss://{container}}@{STORAGE_ACCOUNT}.dfs.core.windows.net"

def read_silver(table_name):
    return spark.read \
        .format("delta") \
        .option(f"fs.azure.account.key.{STORAGE_ACCOUNT}.dfs.core.windows.net", STORAGE_KEY) \
        .load(f"{BASE_URL.format(container='silver')}/{table_name}")

materials_df = read_silver("materials")
inventory_df = read_silver("inventory_snapshot")
sales_orders_df = read_silver("sales_orders")
production_orders_df = read_silver("production_orders")

print("✅ All required Silver tables loaded successfully into memory.")

```

Figure 5.13: Loading Silver Tables

The first Gold table, `fact_inventory_status`, was built to provide a comprehensive view of inventory at the material level. The transformation involved joining the `inventory_snapshot` and `materials` tables, then enriching the result with consumption metrics derived from `production_orders` and sales demand figures from `sales_orders`.

Several business rules were applied during this stage to classify each material record. A column `months_since_consumption` was calculated by dividing the number of days since the last recorded production date by 30. Materials with no consumption history were assigned a value of 999 to ensure they were correctly flagged. The `total_value` field was derived by multiplying `quantity_on_hand` by `unit_cost`. Four classification flags were then applied, as shown in Figure 5.14.

```

# Business Rules
fact_inventory_status = fact_df \
    .withColumn("months_since_consumption",
        F.datediff(F.current_date(), F.col("last_consumption_date")) / 30.0) \
    .withColumn("total_value", F.col("quantity_on_hand") * F.col("unit_cost")) \
    .withColumn("is_recoverable_asset",
        F.when(F.col("quantity_on_hand") > F.col("twelve_month_consumption"), 1).otherwise(0)) \
    .withColumn("is_mc_dead_stock",
        F.when(F.col("months_since_consumption") >= 9, 1).otherwise(0)) \
    .withColumn("provision_amount",
        F.when(F.col("is_mc_dead_stock") == 1, F.col("total_value")).otherwise(0)) \
    .withColumn("stock_deficit", F.lit(0).cast("long")) \
    .withColumn("at_stockout_risk", F.lit(0).cast("integer"))

```

Figure 5.14: Business Rules Applied to Inventory Data

The business rules were defined as follows:

1. `is_recoverable_asset`: Flagged as 1 if `quantity_on_hand` exceeds `twelve_month_consumption`, indicating surplus stock.
2. `is_mc_dead_stock`: Flagged as 1 if the material has not been consumed in 9 or more months.
3. `is_mc_expired`: Flagged as 1 if the material's earliest expiry date has already passed the current date.
4. `provision_amount`: Set to the full `total_value` if the material is either dead stock or expired, otherwise 0.
5. `stock_deficit`: Calculated as the difference between `sales_demand` and `quantity_on_hand` where demand exceeds available stock.
6. `at_stockout_risk`: Flagged as 1 if a `stock_deficit` exists, indicating the material cannot fulfil open orders.

Once all transformations were applied, the resulting DataFrame was written to the Gold container in Delta format using overwrite mode with schema evolution enabled, as shown in Figure 5.14.

```
> consumption_metrics: pyspark.sql.connect.dataframe.DataFrame = [material_id: string, twelve_month_consumption: double ... 1 more field]
> fact_df: pyspark.sql.connect.dataframe.DataFrame = [material_id: string, quantity_on_hand: double ... 9 more fields]
> fact_inventory_status: pyspark.sql.connect.dataframe.DataFrame = [material_id: string, quantity_on_hand: double ... 16 more fields]
> inventory_df_local: pyspark.sql.connect.dataframe.DataFrame = [snapshot_date: string, material_id: string ... 3 more fields]
> materials_df_local: pyspark.sql.connect.dataframe.DataFrame = [material_id: string, material_name: string ... 5 more fields]
> production_orders_df_local: pyspark.sql.connect.dataframe.DataFrame = [production_order_id: string, production_date: string ... 3 more fields]
> raw_inventory: pyspark.sql.connect.dataframe.DataFrame = [material_id: string, quantity_on_hand: double ... 6 more fields]
🚀 fact_inventory_status saved to Gold!
```

Figure 5.14: `fact_inventory_status` Successfully Written to Gold

The second Gold table, `fact_order_fulfillment_risk`, was designed to identify open sales orders that are at risk of not being fulfilled due to material shortages. This table was built on top of the `fact_inventory_status` results computed in the previous step.

The transformation began by filtering `fact_inventory_status` to isolate materials flagged with `at_stockout_risk = 1`. These at-risk materials were then cross-referenced with `production_orders` to establish a mapping between finished products and their constituent materials. This mapping was joined against `sales_orders` filtered to only Open status records, as shown in Figure 5.15.

```

# Get materials at stockout risk
stockout_materials = fact_inventory_status \
    .filter(F.col("at_stockout_risk") == 1) \
    .select("material_id", "stock_deficit")

# Get finished products that use at-risk materials
at_risk_products = production_orders_df_local \
    .select("finished_product", "material_id") \
    .distinct() \
    .join(stockout_materials, "material_id", "inner")

# Join with open sales orders
order_risk = sales_orders_df \
    .filter(F.col("status") == "Open") \
    .join(
        at_risk_products,
        sales_orders_df.finished_product == at_risk_products.finished_product,
        "inner"
    ) \
    .select(
        sales_orders_df.sales_order_id,
        sales_orders_df.order_date,
        sales_orders_df.customer_name,
        sales_orders_df.finished_product,
        sales_orders_df.quantity_ordered,
        sales_orders_df.required_delivery_date,
        at_risk_products.material_id,
        at_risk_products.stock_deficit
    )

```

Figure 5.15: Order Fulfillment Risk Join Logic

The resulting table captured key fields including `sales_order_id`, `customer_name`, `finished_product`, `quantity_ordered`, `required_delivery_date`, `material_id`, and `stock_deficit`. It provides an order-level view of fulfillment exposure. The completed table was then saved to the Gold container in Delta format, as shown in Figure 5.16.

```

> at_risk_products: pyspark.sql.connect.dataframe.DataFrame = [material_id: string, finished_product: string ... 1 more field]
> order_risk: pyspark.sql.connect.dataframe.DataFrame = [sales_order_id: string, order_date: string ... 6 more fields]
> sales_orders_df: pyspark.sql.connect.dataframe.DataFrame = [sales_order_id: string, order_date: string ... 5 more fields]
> stockout_materials: pyspark.sql.connect.dataframe.DataFrame = [material_id: string, stock_deficit: double]
✓ fact_order_fulfillment_risk saved to Gold!

```

Figure 5.16: `fact_order_fulfillment_risk` Successfully Written to Gold

5.6 Synapse Analytics Setup

5.6.1 Serverless SQL Pool Database Configuration

Following the transformation and analytical enrichment of the PPG datasets within the Azure Databricks Medallion architecture, the resulting structured data model needs to be made

accessible for high-performance enterprise BI querying. Azure Synapse Analytics was implemented to provide a high-performance relational abstraction layer over the Data Lake storage zone. Specifically, a Serverless SQL Pool instance was utilized to deploy a logical data warehouse database name ‘ppg_gold_warehouse’. This structural framework eliminates the compute and monetary overhead of maintaining a dedicated, always-on relational database engine, allowing PPG to execute on-demand Transact-SQL (T-SQL) operations directly against storage.

5.6.2 Loading Gold Layer Tables into Synapse

To load and structure the analytical Gold layer data within Synapse Analytics, an orchestration pipeline named ‘Pipeline 1’ was configured and executed within Synapse Studio. The successful completion of this pipeline establishes the formal database objects within the workspace resources explorer. As verified by the system object schema directory, the process compiles the data into an organized relational structure under the database views tree, making `dbo.dim_dates`, `dbo.dim_materials`, `dbo.dim_suppliers`, and the central `dbo.fact_inventory_status` table available for SQL execution and external BI connectivity.

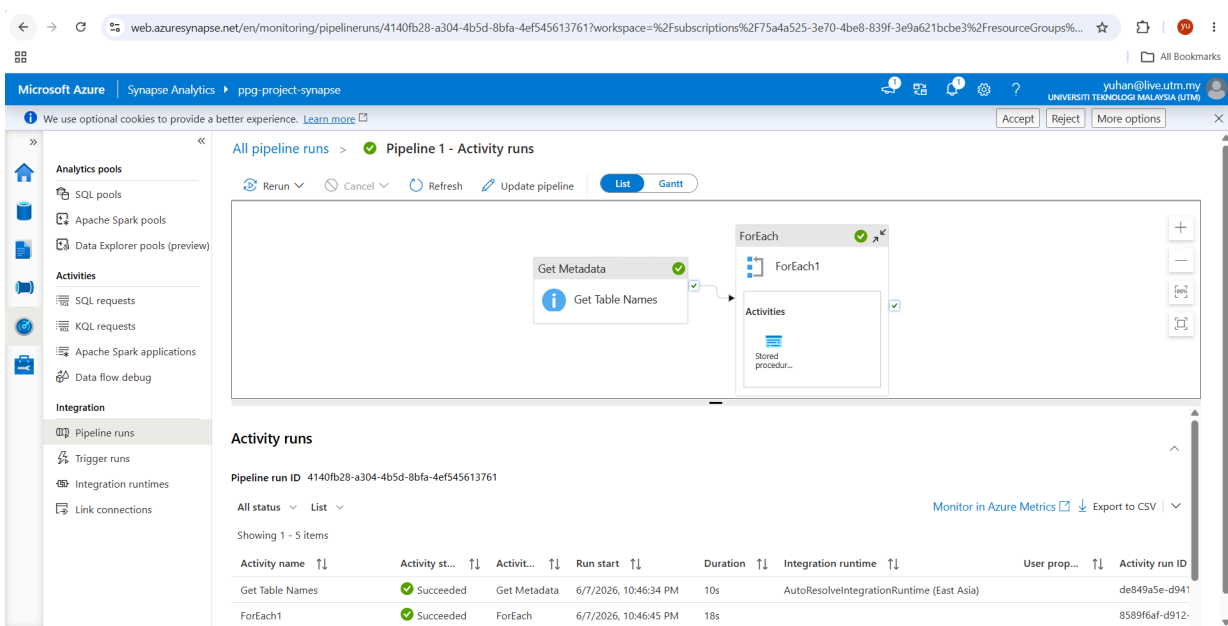


Figure 5.17: Azure Synapse Analytics orchestration monitoring interface showing the successful validation run of Pipeline 1.

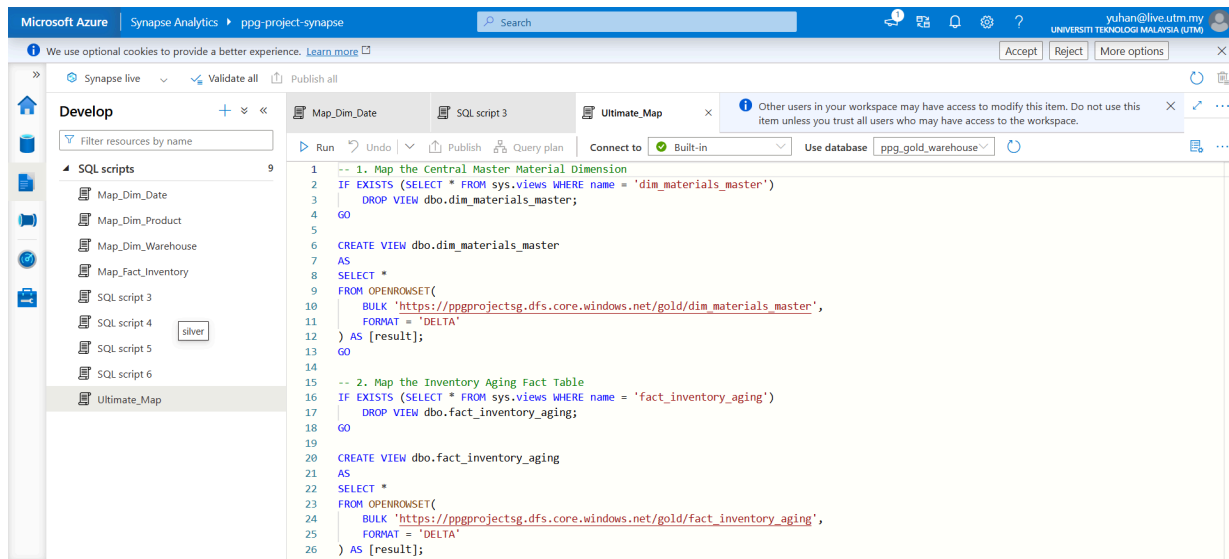


Figure 5.17: Transact-SQL (T-SQL) script deployment inside Synapse Studio orchestrating logical schema creation over Data Lake Delta Objects.

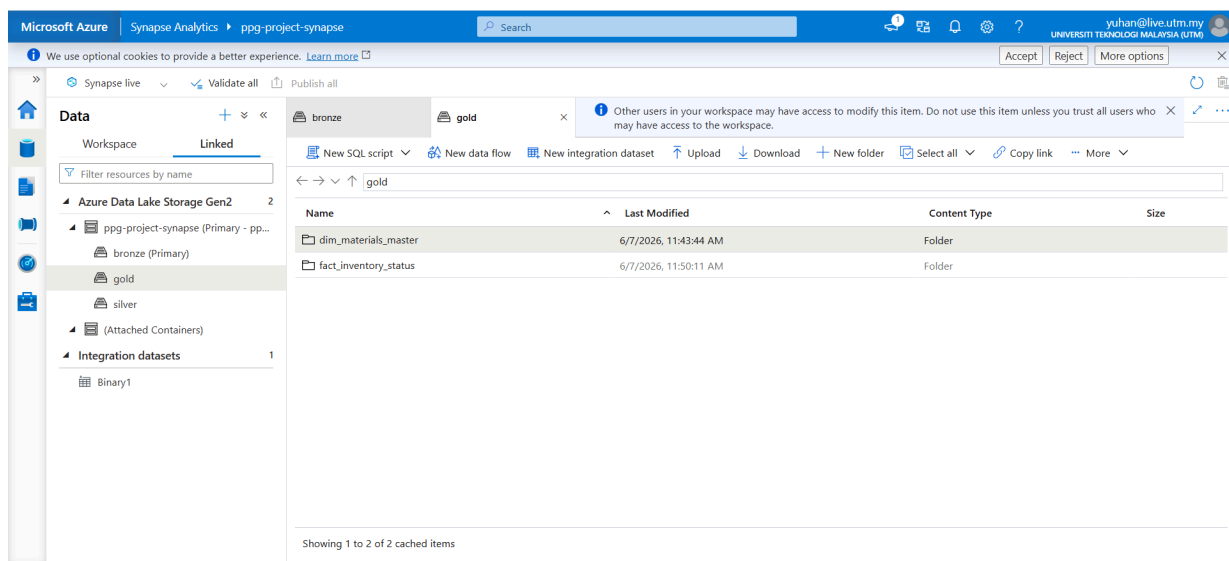


Figure 5.18: Synapse workspace database tree highlighting the compiled external dimensional views ready for downstream BI ingestion

5.6.3 Operational Validation via SQL Querying

In order to verify the analytical integrity of the Star Schema database before exposing the endpoints to the frontend BI dashboard, validation queries were executed within the Synapse Workspace. Analytical test queries successfully isolated specific material metrics, verifying foreign-key consistency between dimension fields and fact evaluations. The execution logs confirm rapid query performance and verified metric computations, validating that the backend data warehouse architecture is stable and ready to serve real-time dashboard visualization layers.

5.7 Power BI Dashboard Implementation

5.7.1 Data Model Connection to Synapse

In Azure Synapse Analytics, all Gold layer tables were created. Then, this group connects the Microsoft Power BI Desktop to the Synapse Serverless SQL Pool endpoint to retrieve the structured star schema data. Figure 5.19 shows the star schema obtained from the gold layer table from the pipeline

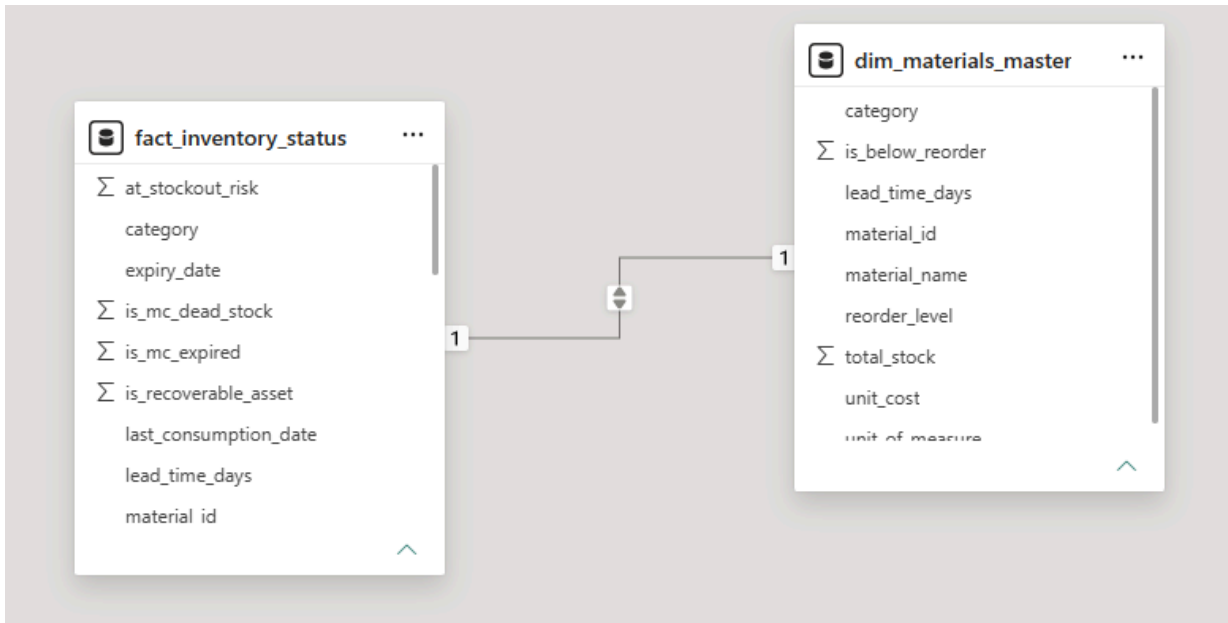


Figure 5.19: Gold Layer Star Schema

Upon importing the two Gold layer tables into Power BI Desktop, the data model relationships were established in the Model view. Power BI automatically detected and configured the relationships as one-to-one (1:1), which reflects the grain of the Gold layer tables. During the Silver-to-Gold transformation in Azure Databricks, both *fact_inventory_status* and *fact_order_fulfillment_risk* were aggregated at the material level, resulting in a single summarized row per *material_id* in each table. As a result, each *material_id* in *dim_materials_master* corresponds to exactly one record in each fact table.

5.7.2 DAX Measures

Custom DAX (Data Analysis Expressions) measures were created to automate the RA and MC business logic calculations and power the KPI scorecards on the dashboard.

```
1 Count Dead Stock Materials =  
2 CALCULATE(  
3 |   COUNTROWS(fact_inventory_status),  
4 |   fact_inventory_status[is_mc_dead_stock] = 1  
5 )
```

Figure 5.20: Count Dead Stock Materials

```
1 Count Expired Materials =  
2 CALCULATE(  
3 |   COUNTROWS(fact_inventory_status),  
4 |   fact_inventory_status[is_mc_expired] = 1  
5 )
```

Figure 5.21: Count Expired Materials

```
1 Count Materials At Stockout Risk =  
2 CALCULATE(  
3 |   COUNTROWS(fact_inventory_status),  
4 |   fact_inventory_status[at_stockout_risk] = 1  
5 )
```

Figure 5.22: Count Materials At Stockout Risk

```

1 Count Materials Requiring Provision =
2 CALCULATE(
3     COUNTROWS(fact_inventory_status),
4     fact_inventory_status[provision_amount] > 0
5 )

```

Figure 5.23: Count Materials Requiring Provision

```

1 Count of Recoverable Materials =
2 CALCULATE(
3     COUNTROWS(fact_inventory_status),
4     fact_inventory_status[is_recoverable_asset] = 1
5 )

```

Figure 5.24: Count of Recoverable Materials

```

1 Total Dead Stock Value =
2 CALCULATE(
3     SUM(fact_inventory_status[total_value]),
4     fact_inventory_status[is_mc_dead_stock] = 1
5 )
6
7

```

Figure 5.25: Total Dead Stock Value

```

1 Total Expired Write-Down =
2 CALCULATE(
3     SUM(fact_inventory_status[provision_amount]),
4     fact_inventory_status[is_mc_expired] = 1
5 )

```

Figure 5.26: Total Expired Write-Down

```

1 Total Materials Below Reorder =
2 CALCULATE(
3     COUNTROWS(fact_inventory_status),
4     fact_inventory_status[quantity_on_hand] < fact_inventory_status[reorder_level]
5 )

```

Figure 5.27: Total Materials Below Reorder

```

1 Total Provision Required =
2 SUM(fact_inventory_status[provision_amount])

```

Figure 5.28: Total Provision Required

```

1 Total RA Value =
2 CALCULATE(
3     SUM(fact_inventory_status[total_value]),
4     fact_inventory_status[is_recoverable_asset] = 1
5 )

```

Figure 5.29: Total RA Value

```

1 Total Stock Deficit =
2 SUM(fact_inventory_status[stock_deficit])
3

```

Figure 5.30: Total Stock Deficit

5.7.3 Dashboard Visualization



Figure 5.31: Materials Below Reorder Level

Figure 5.31 whose current stock has fallen below the designated safety stock threshold. All materials were adequately stocked at the time of reporting. Thus, there is no replenishment action was required

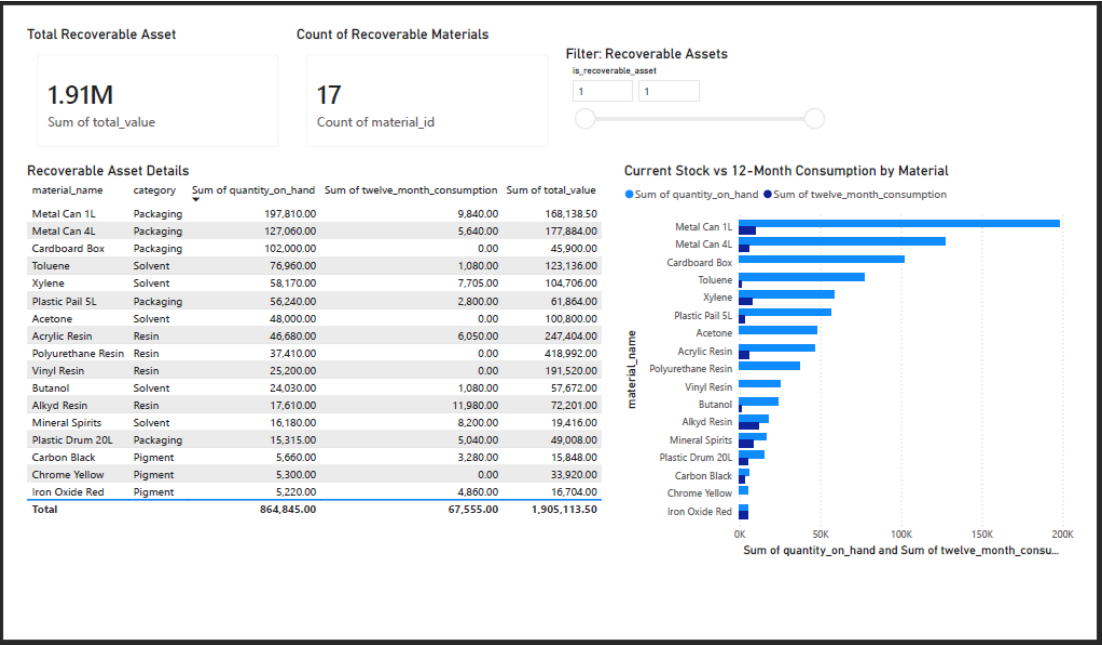


Figure 5.32: Recoverable Assets

Figure 5.32 shows the material flagged as recoverable assets. A bar chart compares current stock quantity against the 12-month consumption forecast per material.

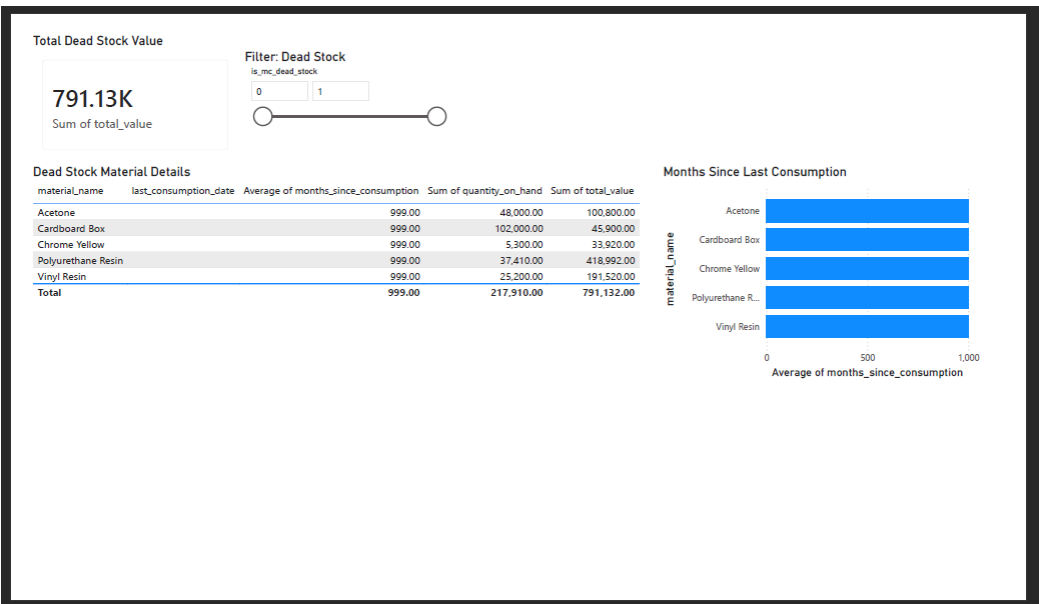


Figure 5.33: Magna Carta Dead Stock

Figure 5.33 shows the materials with zero consumption for 9 or more months totaling RM791132 in at risk value. These materials were assigned a months-since-consumption score of 999, indicating no recorded production activity.

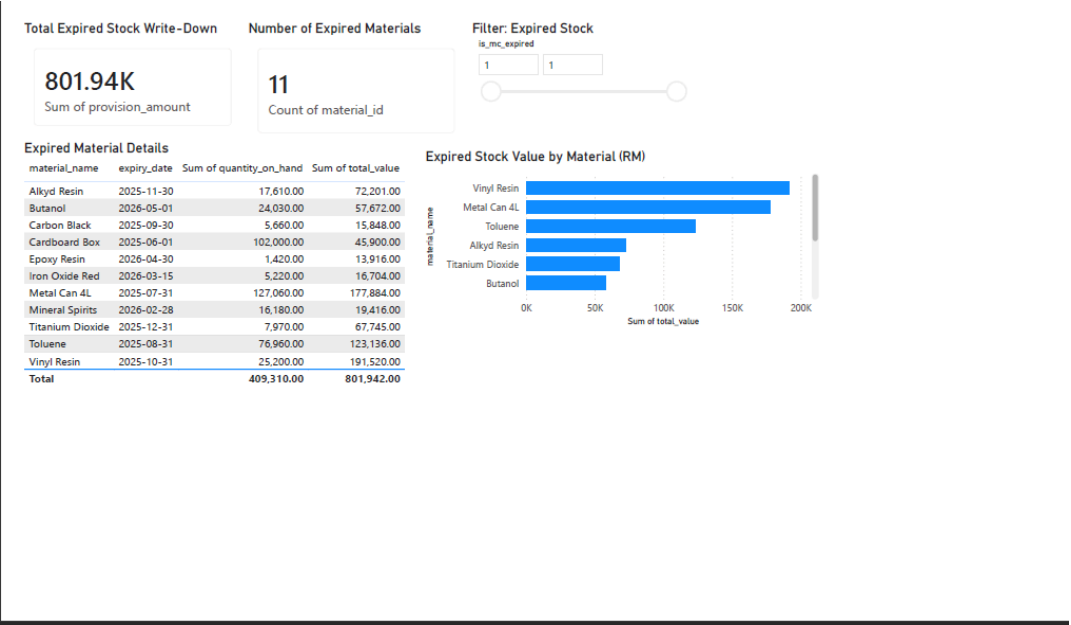


Figure 5.34: Magna Carta Expired Stock

Figure 5.34 shows materials that have passed their certified lot expiry dates, requiring a full 100% financial write-down totalling RM 801,942. The highest-value expired materials were Vinyl Resin (RM 191,520), Metal Can 4L (RM 177,884), and Toluene (RM 123,136).

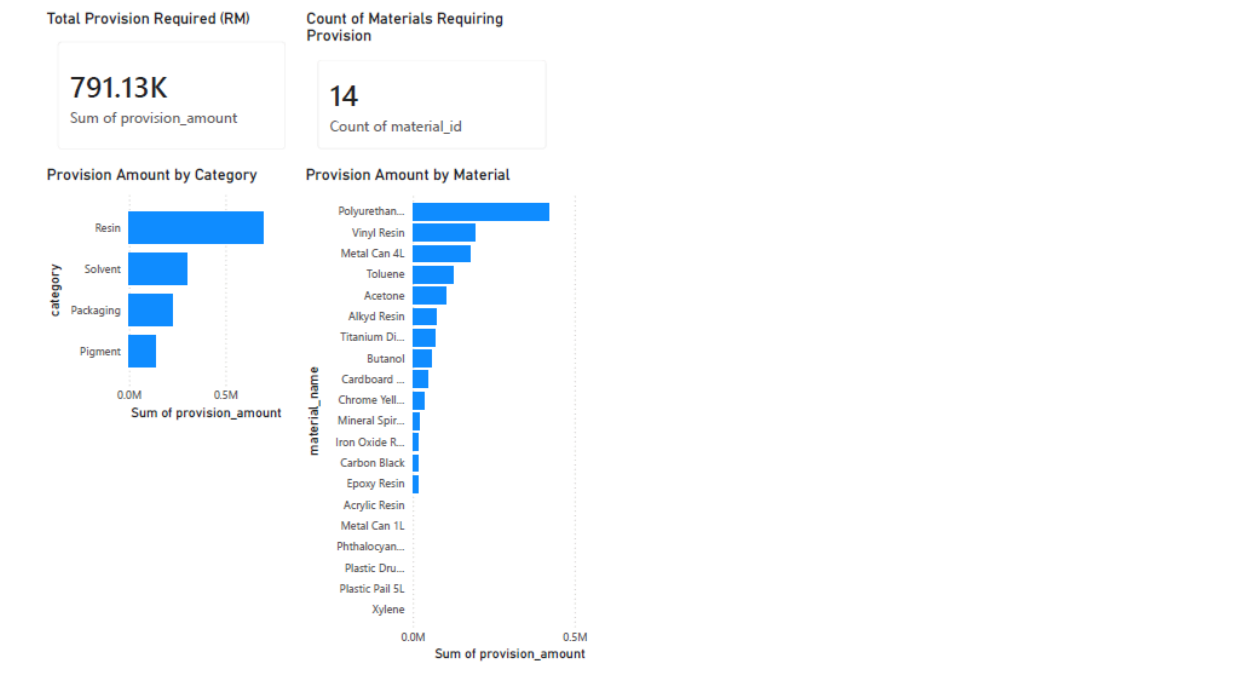


Figure 5.35: MC Provision Required

Figure 5.35 shows the total financial provision liability across all flagged MC materials. 14 materials were identified as requiring financial provision. Resin and Solvent categories carried the highest combined provision burden

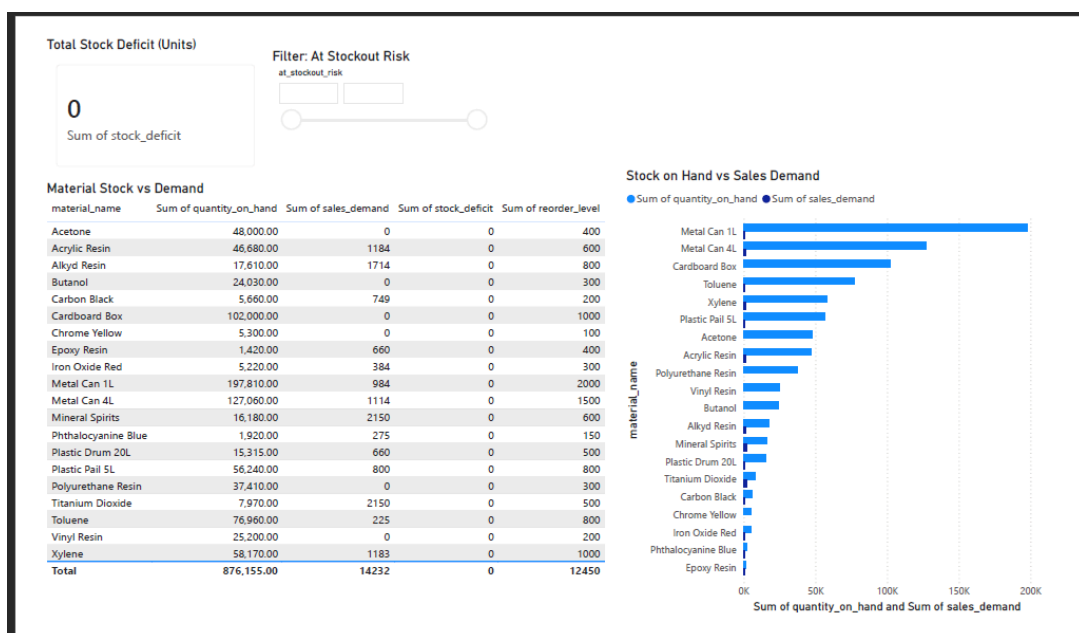


Figure 5.36: Stockout Risk

Figure 5.36 shows the comparison of current stock on hand against open sales demand. The total stock deficit was 0 units, confirming that all 20 materials currently had sufficient stock to cover open sales orders. No materials were flagged at stockout risk at the time of the snapshot.



Figure 5.37: Open Sales Orders at Stockout Risk

Figure 5.37 shows material shortages to specific customer sales orders. Since no stockout risk was detected, no customers were impacted and no sales orders were at risk of fulfilment delay.

5.8 Summary

The technical execution and physical deployment of the PPG Inventory Risk Management analytics pipeline successfully leveraged the Medallion Architecture within the Microsoft Azure ecosystem. By implementing this modern data lakehouse approach, the engineering team transitioned a fragmented, manual data process into a fully automated, scalable cloud solution.

Throughout the implementation phase, each tier of the architecture was systematically constructed: Azure Data Factory reliably orchestrated the ingestion of raw operational files into the Bronze layer; Azure Databricks provided the computational power to cleanse and apply

complex business logic in the Silver layer; and the data was subsequently modeled into a highly optimized Star Schema within the Gold layer. Finally, the integration of Azure Synapse Analytics established a high-performance serving layer, seamlessly bridging the curated data with Microsoft Power BI.

Ultimately, the deployment of this end-to-end infrastructure ensures a secure, reliable, and automated flow of data. It successfully fulfills the project's technical objectives by providing the business intelligence layer with the precise, high-quality metrics required to proactively monitor stock health, calculate financial provisions, and mitigate the risks associated with Recoverable Assets (RA) and Magna Carta (MC) dead stock.

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